

國立陽明交通大學

經營管理研究所

博士論文

Institute of Business and Management

National Yang Ming Chiao Tung University

Doctoral Dissertation

ESG 資訊之揭露對於企業特有風險之影響：

以台灣上市櫃公司為例

Impacts of ESG Information Disclosure on Corporate Idiosyncratic

Risk: Evidence from Taiwan's Listed Companies

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中華民國一一五年一月

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博士論文

A Dissertation Submitted to
Institute of Business and Management
College of Management
National Yang Ming Chiao Tung University
in Partial Fulfillment of the Requirements
for the Degree of
Doctor of Philosophy
in
Business and Management

January 2026

Taiwan, Republic of China

中華民國一一五年一月

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摘 要

近年來，環境、社會與公司治理（ESG）因素逐漸被納入企業永續發展策略與風險管理架構之中，並成為資本市場關注的重要資訊來源。然而，既有實證研究對於各個 ESG 構面，以及 ESG 相關負面事件，究竟如何影響企業風險，仍未形成一致結論。基於此，本論文從企業環境表現、社會責任績效、公司治理特徵，以及 ESG 負面事件揭露等面向出發，系統性探討其與企業特有風險之間的關聯性。根據訊號理論，將永續資訊揭露視為企業向市場傳遞內在品質與治理能力的重要管道，惟正向與負向 ESG 訊息對投資人的企業風險認知之影響，可能有所差異，而將 ESG 負面事件作為風險訊號納入分析的研究仍相對有限。

實證結果顯示：在環境面向中，用水密集度與總二氧化碳排放密集度皆與企業特有風險呈現顯著正向關係，顯示對水資源依賴程度越高或整體碳排放負擔越重之企業具有較高的風險暴露特性。社會面向而言，以職業安全相關指標衡量時，並未觀察到對企業特有風險的顯著影響，顯示社會績效資訊在目前情境下，尚未成為投資人評估企業風險的關鍵依據。於公司治理面向中，董事會規模較大與女性董事比例越高的企業，其特有風險相對較低，顯示董事會監督功能與多元性有助於降低企業特有風險；而在股權結構方面，大股東持股集中度越高，企業所承擔的風險反而越高。此外，無論以發生頻率或事件強度衡量，ESG 負面事件皆未對企業特有風險產生顯著影響。

整體而言，本論文建構一個同時納入環境、社會、公司治理及 ESG 負面事件的多構面分析架構，彌補過去研究多聚焦單一 ESG 構面的限制。研究結果顯示，企業特有風險主要受到公司治理品質與特定環境相關營運風險的影響，而非單純取決於 ESG 揭露的數量或呈現形式。此一發現強調，在進行永續相關風險評估時，應更重視實質內容而非表面揭露，並為理解永續資訊揭露、公司治理機制與投資人風險認知之間的關聯提供實證支持。

關鍵詞：ESG、企業特有風險、訊號理論、公司治理、ESG 負面事件、二階段最小平方法迴歸分析。

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ABSTRACT

In recent years, environmental, social, and governance (ESG) factors have increasingly been incorporated into corporate sustainability strategies and risk management frameworks, becoming important sources of information for capital market participants. However, prior empirical studies have yet to reach a consensus on how individual ESG dimensions, as well as ESG-related negative events, affect enterprise risk. Accordingly, this dissertation systematically investigates the relationship between enterprises' environmental performance, social responsibility outcomes, corporate governance characteristics, ESG-related negative event disclosures, and idiosyncratic risk. Drawing on signaling theory, sustainability disclosures are viewed as an important channel through which enterprises convey their underlying quality and governance capabilities to the market. Nevertheless, the effects of positive and negative ESG signals on investors' risk perceptions may differ, and empirical research that explicitly incorporates ESG-related negative events as risk signals remains relatively limited.

The empirical results indicate that, within the environmental dimension, both water intensity and total carbon emission intensity show significantly positive relationships with idiosyncratic risk, indicating that enterprises with greater dependence on water resources or higher overall carbon emission burdens face higher risk exposure. In the social dimension, occupational safety-related indicators do not exert a significant effect on idiosyncratic risk, suggesting that social performance information has not yet become a key factor in investors' risk assessments under the current context. In the governance dimension, enterprises with larger board sizes and higher proportions of female directors tend to exhibit lower idiosyncratic risk, implying that board monitoring capacity and board

diversity effectively mitigate enterprise risk. Regarding ownership structure, higher ownership concentration among major shareholders is associated with increased enterprise risk. Furthermore, ESG-related negative events, whether measured by occurrence frequency or event severity, do not show a significant impact on idiosyncratic risk.

Overall, this dissertation develops a multi-dimensional analytical framework that simultaneously incorporates environmental, social, governance, and ESG-related negative events, addressing the limitations of prior studies that focus predominantly on single ESG dimensions. The findings indicate that enterprise-specific risk is primarily driven by governance quality and specific environment-related operational risks, rather than by the quantity or form of ESG disclosures alone. This evidence underscores the importance of substance over form in sustainability-related risk assessments and provides empirical support for understanding the linkages among sustainability disclosures, corporate governance mechanisms, and investors' risk perceptions.

Keywords: ESG, Enterprise Idiosyncratic Risk, Signaling Theory, Corporate Governance, ESG-Related Negative Events, Two-Stage Least Squares (2SLS) Regression Analysis.

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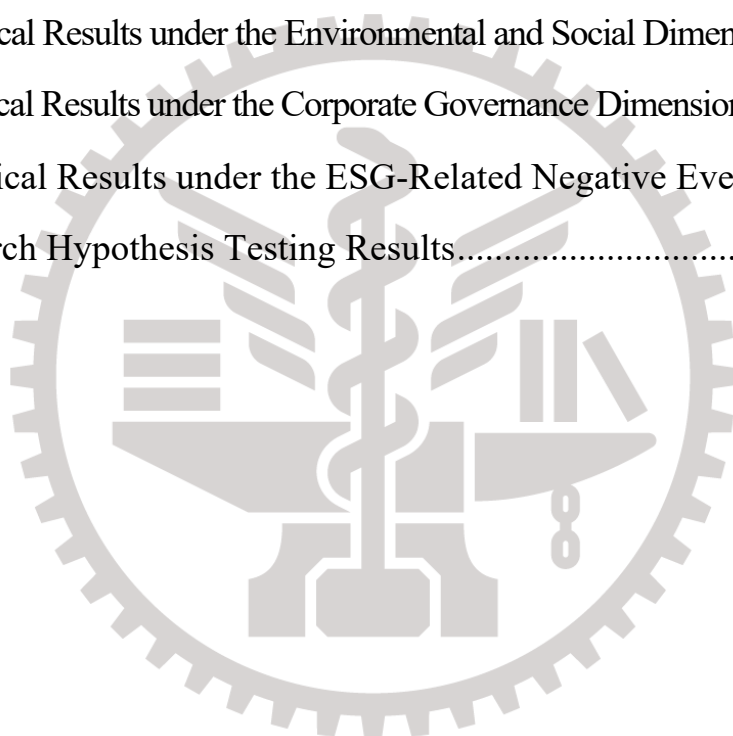
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I. Introduction

1.1 Background

Over the past decade, global systems have been increasingly exposed to large-scale and interconnected disruptions, including public health crises, accelerating climate change, and mounting pressures on natural resources (Daher et al., 2021; Paudel et al., 2023). These developments have reshaped everyday life, economic organization, and social institutions, while simultaneously intensifying the challenges associated with achieving long-term sustainability (George et al., 2016). Because such risks transcend national borders and policy domains, international organizations, governments, and scholars have increasingly emphasized the need for coordinated global responses to reduce environmental degradation, address structural resource inequalities, and safeguard sustainable development pathways (Waddock, 2008). For enterprises, the level of risk directly influences resource allocation, strategic choices, and operational models. Enterprises that are able to accurately identify the sources and magnitude of risk are better positioned to formulate forward-looking strategies (Das & Teng, 1998; Huang et al., 2022; Mariadoss et al., 2014).

The institutional foundations of sustainable development were laid with the Brundtland Commission's definition in the early 1980s, after which sustainability gradually became embedded within global governance arrangements (Kidd, 1992). The widespread adoption of the sustainable development has not only shaped policy agendas but has also redirected academic debate toward a clearer understanding of how enterprises contribute to sustainable development, prompting a noticeable growth in the concepts and tools used to study this relationship (Antolín-López et al., 2016). In this context, supranational institutions, most notably the European Union (EU) and the United Nations (UN), have taken an active role in shaping policy initiatives aimed at responding to these systemic transformations (Barkemeyer et al., 2014; Leite et al., 2025). During expanding body of work, "corporate sustainability" is

frequently positioned as a guiding lens: it frames sustainability as a long-horizon managerial commitment in which business decisions are expected to reflect environmental stewardship and social responsibility, rather than prioritizing immediate financial returns (Bansal, 2005).

This orientation has been reinforced by the rising prominence of ESG thinking in corporate governance and capital markets. Following the United Nations' introduction of ESG principles in 2004, sustainability-related criteria have increasingly been treated as decision-relevant information, influencing how enterprises set strategy and how performance is assessed (Aldowaish et al., 2022). Importantly, the shift has not been driven by corporate initiatives alone. Regulatory intervention, most visibly in the European Union, has strengthened the institutional foundation for sustainable finance. Measures such as the Sustainable Finance Disclosure Regulation and the EU Taxonomy Regulation have helped define what should be disclosed and how “sustainable” activities are classified, thereby moving sustainability governance away from voluntary reporting norms and toward enforceable, rules-based requirements.

As international sustainability frameworks have become more institutionalized, enterprises have increasingly come to view sustainable development not as an auxiliary concern but as a fundamental element of long-term corporate strategy. Whereas earlier sustainability efforts were often motivated by reputational considerations or traditional corporate social responsibility (CSR) objectives (Du et al., 2010), contemporary approaches emphasize the strategic integration of ESG issues with competitiveness, risk management, and stakeholder value creation (George et al., 2023). Stakeholder expectations have likewise evolved, with growing pressure on enterprises to demonstrate credible responses to ESG challenges, reflecting the recognition that durable business performance depends on stable social and environmental systems. Consequently, corporate sustainability has become a focal theme across management, accounting, finance, and sustainability research. This trend is reinforced by the emergence of global sustainability disclosure standards developed by organizations such as

SASB, IIRC, and TCFD, which underscore the growing importance of environmental and social information in investment analysis and corporate risk assessment.

Under the dual pressures of increasingly comprehensive regulatory frameworks and rising stakeholder expectations, enterprises have begun to assume sustainability responsibilities more proactively by engaging in sustainable development initiatives and delivering products and services with long-term value (Matten & Crane, 2005). Corporate efforts toward sustainability-oriented strategies have become increasingly evident, reflecting the growing importance of genuinely embedding sustainability principles into daily operations and decision-making processes (Muñoz-Torres et al., 2019). At the same time, the concept of CSR has continued to evolve, moving beyond a narrow focus on philanthropy or social obligations toward a comprehensive managerial approach that integrates ESG issues. This evolution positions CSR as a governance mechanism that guides enterprises in incorporating sustainability considerations into their core operational frameworks (Bel-Oms, 2025; De Falco et al., 2024; Matten & Moon, 2008).

However, corporate engagement in sustainability is not always as ideal as expected. Some enterprises may respond to sustainability pressures through largely symbolic actions without implementing substantive improvements (Laufer, 2003). More critically, certain enterprises have been involved in severe social or environmental misconduct, causing tangible harm to external stakeholders (Eberl et al., 2015; Schembera & Scherer, 2017). The disclosure of such negative events is often accompanied by information opacity, selective disclosure, or even deliberate concealment, suggesting deficiencies in internal governance mechanisms that hinder enterprises from adequately recognizing and addressing the risks and consequences of harmful practices. Prior studies further indicate that these phenomena reflect insufficient oversight or structural weaknesses in corporate governance systems when enterprises confront social and environmental challenges (Bazerman & Tenbrunsel, 2011; Greve et al., 2010; Scherer et al., 2013).

1.2 Motivation

Early discussions of corporate governance were largely grounded in agency theory, emphasizing the need to address conflicts of interest that arise when ownership and control are separated. Within this framework, governance mechanisms were designed mainly to constrain managerial discretion and safeguard shareholders from self-interested behavior. More recently, this narrow, finance-oriented understanding has been progressively challenged, as corporate governance is increasingly expected to account for enterprises' responsibilities toward a wider set of social and environmental concerns (Aguilera et al., 2007; Shleifer & Vishny, 1997). These sustainability-related governance concerns largely stem from increasing stakeholder expectations regarding corporate responsibility and transparency (Aureli et al., 2020; Mason, 2020).

As sustainability has gained prominence on the global agenda, it has progressively shifted from a peripheral concern to a core element of corporate strategy and value chain configuration (Iansiti & Levien, 2004; Vermeulen & Seuring, 2009). In response to this shift, both regulators and academic scholars have begun to treat corporate governance and sustainability as mutually reinforcing dimensions of organizational oversight, rather than as separate domains. Within this evolving perspective, ESG has risen as a pivotal analytical lens in international corporate governance debates (Eccles et al., 2014; Elamer & Boulhaga, 2024). Importantly, ESG frameworks extend beyond the traditional governance focus on agency cost reduction, placing greater emphasis on enterprises' integrated performance in environmental stewardship, social accountability, and governance effectiveness (Elamer & Boulhaga, 2024). By adopting such a holistic orientation, organizations are better positioned to address stakeholder expectations while simultaneously reducing the adverse social and environmental consequences of their operational activities (Karwowski & Raulinajtys-Grzybek, 2021).

The concept of ESG originates from the foundational principles of traditional corporate governance (Rossi & Tanweer, 2023). Its core objective is to establish institutional and organizational arrangements that ensure enterprises operate appropriately while balancing the interests of multiple stakeholders, whose interests may often conflict, and simultaneously addressing non-financial responsibilities related to social and environmental concerns. The scope of ESG is inherently broad, encompassing various aspects of corporate operations. It includes not only enterprises' compliance with specific rules, standards, and regulations, but also the design and implementation of internal governance structures and managerial processes (Müller et al., 2025; Li et al., 2024). The sources of ESG-related norms extend beyond national legal frameworks and policy requirements to include internally developed managerial procedures and control mechanisms within enterprises (Scherer et al., 2016; Singhania & Saini, 2023). Accordingly, ESG can be viewed as an overarching framework that guides corporate decision-making, with the ultimate aim of steering enterprises toward sustainable development. Within this framework, corporate governance serves as the central mechanism through which this vision is realized. In other words, ESG not only defines the principles and standards that enterprises are expected to follow, but also delineates the pathways for achieving sustainability objectives and the criteria by which outcomes are evaluated.

Moreover, the meaning and scope of sustainability have gradually evolved over time. In its early stages, environmental impacts were often regarded as part of CSR, primarily framed as issues of legal compliance or moral obligation and largely detached from enterprises' core business models and market demands (Berger-Walliser & Scot, 2018). Over time, however, a range of factors has prompted enterprises to reassess sustainability issues and to increase their engagement in sustainable practices. These factors include heightened corporate awareness of sustainability goals, growing pressure from more stringent regulatory requirements, the need to balance product quality with cost efficiency, rising consumer expectations regarding environmental responsibility, and the emergence of new business opportunities associated with

sustainability (Poddar et al., 2019; Weyzig, 2009). As these drivers have accumulated, enterprises have increasingly recognized the close linkage between environmentally friendly strategies, sustainability-oriented practices, and long-term corporate performance (Fernando et al., 2019). Consequently, sustainability has expanded beyond a narrowly defined environmental concern to become a core element of enterprises' overall business models (Bice, 2017).

While the notion of sustainability has existed for decades, its strategic and societal salience has intensified significantly in recent times. Fundamentally, sustainability embodies a forward-looking perspective that seeks to reconcile current human welfare with the responsibility to safeguard future generations' capacity to satisfy their needs and maintain a satisfactory quality of life (Almagtome et al., 2020; Kjell, 2011). In contemporary organizational discourse, these core ideas are increasingly operationalized through the ESG framework, which gives rise to the concept of "triple sustainability" by jointly addressing environmental integrity, social fairness, and economic viability within a coherent analytical structure (Opp & Saunders, 2013). As stakeholder expectations continue to shift and institutional pressures become more complex, sustainability is expected to move beyond its current boundaries, prompting more refined and multidimensional inquiries across an expanding range of ESG-related domains.

Corporate governance frameworks centered on sustainability have brought about significant changes in enterprises' decision-making and monitoring mechanisms. Enterprises are now required not only to enhance the transparency of their governance structures but also to more effectively communicate information related to social responsibility and stakeholder engagement. As the external environment evolves and information demands increase, accounting and disclosure systems must also adapt by expanding the scope of disclosed information, thereby enabling stakeholders to access sustainability-related information that is more relevant and decision-useful.

Enterprises typically disclose their ESG initiatives through sustainability reports or annual reports. However, such disclosures tend to emphasize positive information, while negative

information is often downplayed or omitted due to reputational concerns. As a result, external evaluations of enterprises' ESG performance frequently rely on self-disclosed information that presents a predominantly favorable narrative. In contrast, news media often reveal negative events that enterprises fail to disclose or deliberately understate, leading investors to perceive ESG-related risks that diverge from the image portrayed by enterprises themselves.

Several high-profile cases illustrate this phenomenon. For instance, following the disclosure of a water pollution incident involving ASE Group in 2013, the enterprise's stock price declined by more than 7% within three days. Similarly, the Ting Hsin oil scandal in 2014 triggered widespread consumer boycotts, resulting in a nearly 30% reduction in market capitalization within one month. These cases demonstrate that ESG-related negative events tend to exert a substantially stronger impact on financial markets than the voluntarily disclosed positive information, and they can rapidly alter investors' perceptions of corporate risk.

Accordingly, this dissertation quantitatively examines media-reported ESG-related negative events to investigate their substantive effects on investors' risk assessments. By simultaneously incorporating both positive and negative information, this research aims to provide a more comprehensive understanding of the true effectiveness of corporate sustainability strategies and to uncover the potential trade-offs and latent risks that enterprises may face in the process of pursuing sustainable development.

1.3 Research Objectives

Existing research has largely emphasized the beneficial consequences of ESG engagement for enterprises, forming a substantial and cumulative body of evidence. This literature consistently highlights ESG performance as a driver of more sustainable governance structures, enhanced corporate legitimacy, and improved financial outcomes. For example, Halid et al. (2023) report that organizations with stronger ESG ratings tend to achieve better operational results while facing lower levels of enterprise-specific risk, indicating that ESG initiatives can

function as an effective risk-buffering mechanism. Similarly, Khan (2022) employs a random-effects estimation approach and finds that proactive involvement in ESG-related activities is associated with superior financial performance. Extending beyond financial metrics, Cabaleiro-Cerviño and Mendi (2024), drawing on stakeholder theory and social network innovation theory, show that ESG adoption fosters enterprises' future-oriented innovation capabilities. Moreover, evidence from periods of economic turbulence suggests that ESG and corporate governance practices may play a strategic role in sustaining enterprise performance under conditions of elevated uncertainty (Ahmad et al., 2023).

In contrast, how ESG-related negative information influences enterprise risk and market valuation remains fragmented and lacks a systematic analytical framework. Moreover, much of the existing literature relies on composite ESG ratings as an overall measure, paying relatively limited attention to specific indicators within the ESG dimensions. This aggregation makes it difficult to identify which sustainability issues stakeholders are most concerned about and whether different ESG dimensions exert heterogeneous effects on enterprise risk. Consequently, there remains a need for a more granular examination of ESG components and negative events to better understand their distinct roles in shaping corporate risk and investor perceptions.

Building on the preceding discussion, this dissertation seeks to construct a more refined and actionable framework for assessing sustainability by jointly considering ESG-based sustainability indicators and internal corporate governance attributes, including board structure and ownership arrangements. Rather than treating sustainability as a composite construct, the study explicitly investigates how these governance and sustainability-related factors shape enterprises' exposure to idiosyncratic risk. To overcome the limitations of prior studies that rely heavily on aggregate ESG ratings, this dissertation incorporates disaggregated and outcome-oriented indicators, such as carbon and water intensity, occupational injury incidence, and the occurrence of ESG-related adverse events. Via this approach, the analysis captures the differential risk implications of distinct sustainability dimensions, thereby offering a more

detailed account of how specific ESG practices are translated into heterogeneous risk profiles across enterprises.

At the same time, this dissertation seeks to bridge the gap between sustainability disclosures and market reactions by examining whether enterprises' disclosed sustainability performance (positive information) and ESG-related negative events (negative information) are reflected in investors' assessments of enterprise risk. In doing so, the dissertation provides a theoretical framework to explain how sustainability-related information is transmitted to and interpreted by external capital markets. Through this approach, the dissertation not only aims to clarify the mechanisms through which positive and negative ESG-related information influences enterprise risk, but also aspires to offer practical implications for sustainability disclosure practices and regulatory design in Taiwan. Moreover, the findings are intended to lay a foundation for future research at the intersection of sustainability, corporate governance, and enterprise risk.

1.4 Structure of This Dissertation

The dissertation is presented in five chapters, each serving a distinct purpose. Chapter I introduces the study by describing the research setting and the key problem that motivates the analysis. It also states the research aims, specifies the scope of the work, and clarifies how the study is positioned. Chapter II reviews the core literature on ESG and enterprise risk and summarizes the theoretical arguments that underpin the research questions. Drawing on this discussion, it develops hypotheses for each ESG pillar, environmental, social, and governance, and extends the hypothesis development to include ESG-related adverse event disclosures. Chapter III explains how the empirical investigation is carried out. It outlines the research design in detail, including variable definitions and measurement procedures, data sources, data period selection, and the econometric models employed. To potential endogeneity is addressed through two-stage least squares (2SLS) estimation. Chapter IV presents the empirical results

and provides interpretation by connecting the findings to the proposed hypotheses and to prior empirical evidence. Chapter V concludes the dissertation by summarizing the main findings and contributions, discussing managerial implications, acknowledging limitations, and proposing directions for future research.



II. Literature Review and Hypothesis Construction

2.1 The Concept of ESG

In many contexts, the public sector advances sustainability by setting rules and incentives through policy design and regulatory systems, whereas businesses are central to execution, reconfiguring production and operations, expanding sustainable goods and services, and advancing innovation. The European Green Deal introduced by the European Union in 2019 illustrates how sustainability priorities can be embedded directly within governance and policy architectures (Câmara & Morais, 2022). Consistent with the UN sustainability agenda, such policy initiatives reinforce a long-term value creation logic and encourage enterprises to look beyond short-run profit targets toward a broader balancing of stakeholder interests (Hart and Milstein, 2003).

Within this broader agenda, the corporate domain has received sustained attention because organizational choices can accelerate, or hinder, sustainability transitions. Prior research often treats corporate governance arrangements, corporate social responsibility (CSR), and innovation capability as key enablers of progress toward sustainable development objectives (Aboud & Diab, 2019; Moussa et al., 2022). Corporations also shape the landscape of responsible investment by influencing what information is disclosed and how ethical, environmental, and social standards are incorporated into managerial routines. Enterprises that take sustainability commitments seriously tend to embed sustainability priorities into their core business models and increasingly regard sustainability performance as a dimension of competitive positioning (Helfaya & Moussa, 2017).

At the same time, corporate decision-making is increasingly understood as a multidimensional optimization process in which financial objectives must be balanced with environmental stewardship and social obligations (Ellili, 2022; Siegrist et al., 2020). Within this context, corporate boards play a pivotal role by implementing governance arrangements

that explicitly consider the long-term economic, social, and ecological implications of organizational choices. Prior research has examined the design and effectiveness of such governance mechanisms across a variety of industries and institutional environments, highlighting their relevance for sustainable corporate conduct (Aras et al., 2018; Bose et al., 2018; Ellili, 2022; Karaman et al., 2020; Ong & Djajadikerta, 2020).

In the contemporary business landscape, sustainability is no longer viewed as a peripheral managerial concern but has become embedded within the core logic of corporate governance. The concept is commonly articulated through three interconnected dimensions: economic soundness, social responsibility, and environmental preservation (James, 2015; Kocmanová et al., 2020). As this perspective has gained traction, enterprises across sectors have increasingly incorporated sustainability considerations into governance practices, reflecting its fundamentally intergenerational orientation toward safeguarding future resources and forms of capital (James, 2015; López et al., 2007; Cairns, 2006; Gray, 2002). At the organizational level, these commitments are most clearly manifested through structured sustainability reporting and the systematic oversight of ESG-related issues (Oprean-Stan et al., 2020).

Corporate sustainability can therefore be viewed as an evolution of contemporary governance thinking. Unlike conventional governance approaches that prioritize immediate profitability and production efficiency, sustainable governance places stronger weight on long-term growth, value creation, and improvement in organizational effectiveness. It shifts attention away from narrow financial outcomes and toward an enterprise's capacity to generate durable value while meeting environmental and social obligations. At its core, corporate sustainability requires aligning social and environmental objectives with financial goals, rather than treating them as separate agendas. Through transparent disclosures and more comprehensive reporting systems, enterprises communicate legitimacy and demonstrate a sustained commitment to stakeholders over time (Gao & Zhang, 2006).

Accordingly, as a central managerial mechanism within corporate operations, corporate governance primarily aims to promote the stable functioning of capital markets and to facilitate sustained investment inflows (Elhabib, 2024; Talamo, 2011). A robust corporate governance system, both in terms of institutional design and practical implementation, must operate within existing legal and regulatory frameworks while establishing optimal management mechanisms that enhance enterprise value and simultaneously safeguard investor returns and stakeholder interests. Notably, the notion of enterprise value is no longer confined to financial profitability alone; rather, it has gradually expanded to encompass non-financial objectives and sustainability considerations, which have become key priorities for modern corporations.

If corporate governance is viewed merely as a mechanism for supervising board responsibilities, improving board efficiency, or ensuring regulatory compliance, its role becomes overly restrictive. Consequently, sustainability is increasingly regarded as an indispensable component of corporate governance, as the principles of sustainability are inherently intertwined with environmental, social, and economic development. These dimensions likewise constitute integral and inseparable elements of effective corporate governance (Kocmanová et al., 2020).

In recent years, enterprises have faced increasing pressure to disclose non-financial information that affects their value creation processes. Sustainability reporting not only assists enterprises in formulating sustainability-related objectives, but also helps evaluate implementation effectiveness and monitor progress, thereby enhancing enterprises' sustainable management capabilities (Oprean-Stan et al., 2020; Shad et al., 2019). Among various forms of non-financial disclosure, ESG information is particularly critical, as it responds to stakeholders' growing expectations regarding enterprises' sustainability commitments and performance (Fernandez et al., 2019; Tamimi & Sebastianelli, 2017). The disclosure of ESG information facilitates more effective communication between enterprises and their stakeholders and serves as an important basis for governance strategy formulation and

operational decision-making. In the contemporary business environment, enterprises must assess not only traditional financial risks but also ESG-related risks in order to cope with rapid external changes and increasing stakeholder demands (Ashwin Kumar et al., 2016; Oliver Yébenes, 2024; Teng et al., 2021).

In summary, research on sustainability has predominantly centered on the three core dimensions of ESG. Within the environmental dimension, existing studies have examined a wide range of issues, including biodiversity conservation (Kopnina et al., 2024; Pokharel, 2025; Zhu & Carrasco, 2025), climate change (Chen et al., 2022; Li et al., 2024; Oliver Yébenes, 2024), energy consumption (Widianingsih et al., 2024; Yuan et al., 2025), and pollution emissions (Baratta et al., 2023; He et al., 2023; Li & Xu, 2024; Zhang et al., 2023). Research on the social dimension has focused on topics such as human rights protection (Rivlin, 2024), community engagement (Raju & Marisetty, 2025), stakeholder interaction (Cornell & Shapiro, 2021), occupational safety (Jain et al., 2024), employee engagement, customer satisfaction (Seok et al., 2024), gender equality (García-Meca & Martínez-Ferrero, 2025), and diversity and inclusion policies. With respect to the governance dimension, prior studies have primarily emphasized board and managerial composition and competence, executive compensation systems, information transparency and disclosure quality, audit mechanisms, and political contributions.

At the corporate governance level, effective governance systems can enhance enterprises' sustainability capabilities through incentive and monitoring mechanisms. Aguilera et al. (2021) further argue that key governance actors play a critical role in shaping enterprises' sustainability outcomes. Related research has also explored how sustainability-oriented governance frameworks improve the quality of materiality disclosure (MDQ) in international sustainability reporting (Velte, 2022). Accordingly, corporate governance serves not only as the foundation of corporate management but also as an indispensable core dimension within ESG research.

2.2 The Concept of Idiosyncratic Risk

Enterprises inevitably face various types of risks arising from both external environments and internal managerial activities. With rapid changes in global markets and increasing uncertainty, the complexity and likelihood of these risks have continued to intensify. Under such conditions, maintaining a high level of information transparency in investment and managerial decision-making can help mitigate enterprise risk and enhance investor protection. Risk management has become a fundamental responsibility of corporate managers and a central concern shared by both enterprises and investors. Effective risk management not only reflects an enterprise's sound risk awareness but also contributes to greater operational stability and improved decision-making quality, thereby strengthening investor confidence and shaping more favorable risk assessments of the enterprise (Mupa et al., 2024).

Effective risk management also helps mitigate adverse shocks, enabling enterprises to sustain profitability and competitiveness under highly uncertain conditions, thereby strengthening their overall organizational resilience. From the perspective of investors, enterprise risk constitutes an indispensable factor in the valuation of corporate value. The level of risk directly affects the discount rate applied to future cash flows, thereby altering enterprise valuation (Gregory et al., 2014).

Higher risk amplifies uncertainty over future cash flows, thereby inducing more cautious investment decisions (Hirth & Viswanatha, 2011). Consequently, when making asset allocation decisions, investors must assess whether the level of enterprise risk is compensated by commensurate expected returns. If an enterprise exhibits excessively high risk without sufficient growth potential, investors are more likely to reduce their holdings or exit the market altogether.

Enterprise risk is generally divided into systematic and idiosyncratic risk (Bali et al., 2005). Total enterprise risk captures the overall uncertainty in profitability arising from internal

enterprise characteristics and external conditions, and is commonly measured using stock price volatility (Bartram et al., 2015; Luo & Bhattacharya, 2009; Mittnik, et al., 2015). Systematic risk originates from macroeconomic conditions such as economic cycles, fiscal policies, and monetary policies. As a market-wide risk, it cannot be fully eliminated through diversification strategies (Fu, 2009). In contrast, idiosyncratic risk stems from enterprise-specific factors, including internal governance structures, managerial capabilities, corporate culture, and industry characteristics. Such risk can be mitigated through improvements in internal control systems, enhancements in governance mechanisms, and greater information transparency (Gupta et al., 2011). Similarly, Bushee and Noe (2000) and Shahzad et al. (2020) argue that higher levels of information disclosure help reduce investors' expectations of enterprise-level volatility.

Because idiosyncratic risk primarily originates from internal enterprise-level factors, such risk can be mitigated through the strengthening of internal control systems and the enhancement of corporate governance mechanisms. The ESG dimensions are inherently integral components of enterprises' governance and risk management frameworks. Their effective implementation helps constrain managerial behavior, enhance information transparency, and thereby exert a risk-averse effect that reduces idiosyncratic risk. Accordingly, this dissertation focuses on idiosyncratic risk as its core research subject and systematically examines the impact of ESG dimensions on such risk.

This dissertation adopts idiosyncratic risk as the risk measurement variable for two main reasons. First, compared with systematic risk or total risk, idiosyncratic risk is an enterprise-level risk that arises from variations in individual enterprises' operational decisions, governance quality, and internal management factors. As such, it more directly captures the extent to which ESG mechanisms mitigate the unique risks enterprises face during their operational processes. Second, idiosyncratic risk plays a critical role not only in stock market pricing and forecasting

but also in research on portfolio construction and hedging strategies, thereby possessing substantial theoretical and practical significance.

A growing stream of sustainability research treats ESG outcomes as risk-relevant enterprise attributes. Empirical evidence frequently suggests that enterprises receiving stronger ESG evaluations are exposed to less enterprise-specific risk and are less prone to extreme downside outcomes (Lee & Koh, 2024; Reber et al., 2022). Related work reports that better ESG performance is associated with lower market uncertainty, for example, reduced volatility, and with diminished downside tail risk (Reber et al., 2022; Zhang et al., 2021). The governance component is often highlighted in this context. Studies argue that effective governance arrangements can curb internal control weaknesses and discourage corporate misconduct, which in turn helps contain operational risk (Sassen et al., 2016). Consistent with this view, Zhang et al. (2021) show that high-ESG enterprises tend to be more resilient when macroeconomic conditions deteriorate, implying that sustainability practices can reinforce risk management.

Evidence is also strong that governance quality itself is a major determinant of enterprise risk. Under agency theory, robust governance structures can limit managerial opportunism and reduce agency-related costs, thereby lowering the risk ultimately borne by shareholders (Ashbaugh-Skaife et al., 2006; Ho et al., 2013). When governance is weak, the likelihood of governance breakdowns increases. Such weaknesses may translate into higher borrowing costs, incentives toward greater leverage, and, consequently, elevated financial risk. Put differently, enterprises with poorer governance are generally expected to exhibit higher overall risk. Overall, the level of enterprise risk not only affects operational stability, strategic decision-making, and the cost of capital, but also directly shapes investors' assessments of enterprise value and future growth potential.

Systematic risk reflects the macroeconomic and market conditions in which enterprises operate, whereas idiosyncratic risk captures the quality of internal governance and the degree

of information transparency. When enterprises effectively reduce risk by strengthening governance mechanisms, enhancing disclosure practices, and implementing sustainability initiatives, they can improve market resilience and foster greater investor confidence. Therefore, the ability to properly identify and manage both systematic and idiosyncratic risks is of critical and indispensable importance to both enterprises and investors.

2.3 The Relationship between ESG and Idiosyncratic Risk

According to the research framework illustrated in Figure 1, this dissertation examines the effects of the environmental, social, and governance (ESG) dimensions on enterprises' idiosyncratic risk. The ESG-related information is primarily derived from enterprises' sustainability reports and annual reports. However, enterprises tend to disclose a greater amount of favorable information in their official disclosures, while unfavorable or negative events may not be fully revealed due to reputational concerns or risk considerations.

Consequently, such negative information is often difficult to obtain from formal disclosure documents. To address this limitation, this dissertation further collects information on ESG-related negative events from mass media sources and incorporates negative ESG event disclosures into the analysis, thereby compensating for the insufficiency of negative information disclosure in sustainability reports and annual reports. By jointly considering both positive disclosures and negative event information, this dissertation enables a more comprehensive assessment of the overall effects of positive and negative ESG factors on idiosyncratic risk and provides deeper insights into the relationship between ESG performance and corporate risk. Integrating the environmental, social, governance, and ESG negative event dimensions, this dissertation systematically constructs an analytical framework to examine the impact of ESG factors on idiosyncratic risk.

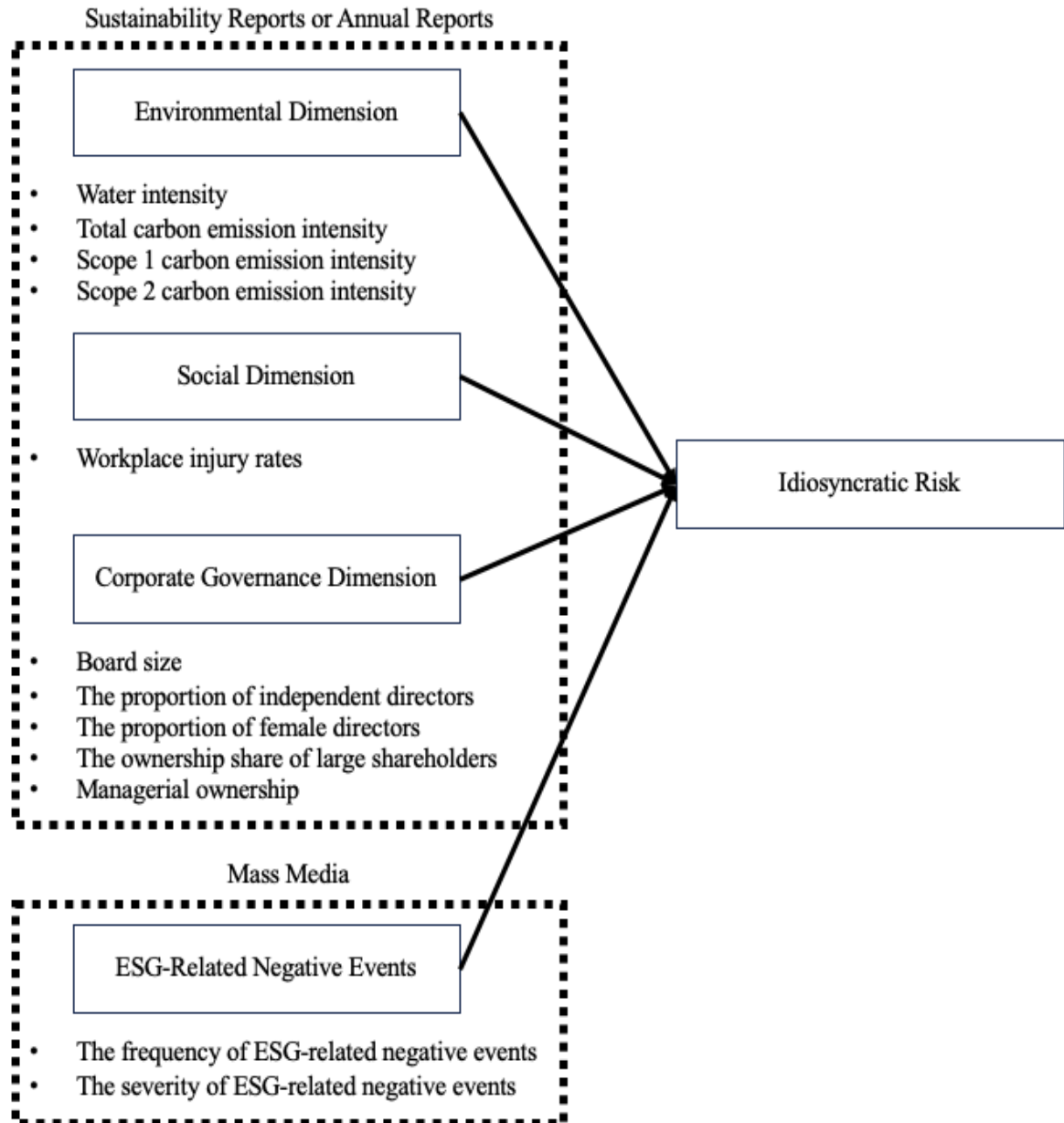


Figure 1. Research Framework

Within capital markets, corporate disclosures such as financial statements and annual reports constitute the primary means through which managers communicate information regarding enterprise performance and governance practices to external investors (Healy & Palepu, 2001; Nandi & Ghosh, 2012). Enhanced disclosure transparency facilitates investor confidence and supports efficient capital allocation, whereas information asymmetry heightens uncertainty and amplifies perceived risk. As a result, investors generally exhibit a preference for enterprises with strong financial fundamentals and more credible disclosure practices. By

broadening both the depth and reliability of disclosed information, managers enable market participants to form a more accurate and comprehensive assessment of the enterprise, thereby alleviating information asymmetry (Botosan, 1997; Glosten & Milgrom, 1985). In this regard, high-quality disclosure improves the functioning of information transmission, enhances valuation precision, and contributes to lower transaction costs in capital markets (Bushman & Smith, 2003).

Prior research has largely relied on agency theory and information asymmetry theory to explain the link between corporate social responsibility and systematic risk (Beyer & Guttman, 2012; Cai et al., 2016). These perspectives typically view disclosure decisions as outcomes of rational, economically motivated behavior. However, they devote relatively limited attention to the possibility that disclosure practices themselves may shape managerial conduct through signaling mechanisms. Signaling theory offers a complementary lens by conceptualizing sustainability-related disclosure not only as a vehicle for information provision but also as a strategic signal directed toward stakeholders. From the standpoint of ESG signal transmission and subsequent market responses, signaling theory thus provides an alternative explanatory pathway through which sustainability disclosure may influence enterprises' exposure to systematic risk.

In corporate disclosure practices, information can generally be classified into positive and negative messages (Groening & Kanuri, 2013). Enterprises, however, tend to disclose information that enhances corporate image or enterprise value, resulting in sustainability reports that predominantly emphasize performance achievements, social contributions, and sustainability accomplishments. Chuprinin et al. (2015) demonstrate that market reactions to positive and negative information are asymmetric: investors exhibit a stronger preference for enterprises disclosing positive information, and portfolios composed of such enterprises tend to generate higher returns. This finding suggests that investors frequently regard positive ESG disclosures as an important input in investment decision-making.

A growing body of evidence suggests that institutional investors tend to direct capital toward enterprises demonstrating stronger social and environmental performance (Dyck et al., 2019; Hong & Kacperczyk, 2009; Kordsachia et al., 2022; Nofsinger et al., 2019). Related research further indicates that active engagement in corporate social responsibility initiatives is associated with improved financial outcomes, greater organizational stability, and lower overall risk exposure (Blasi et al., 2018; Cho et al., 2019; Kuo et al., 2021). Viewed collectively, these findings imply that the transparent communication of favorable ESG practices plays an important role in reinforcing investor trust and easing enterprises' access to external capital.

Compared with positive information, investors tend to interpret negative news as an early warning signal of potential corporate losses, as such information may indicate contractions in operations, unstable cash flows, and ultimately a decline in enterprise value (Tetlock, 2007). Once a major negative event is disclosed, financial markets typically react rapidly, often resulting in sharp declines in stock prices. More importantly, the impact of negative information is frequently stronger than that of positive disclosures. Because sustainability-related information is largely disclosed on a voluntary basis, investors often face difficulties in directly observing enterprises' actual sustainability practices (Aureli et al., 2020; Carmo & Miguéis, 2022). Moreover, disclosed information may be subject to overly optimistic framing or selective presentation (Jo & Na, 2012). Some enterprises may even deliberately omit unfavorable social or environmental performance information (Stacchezzini et al., 2016) or treat sustainability disclosure as a reputation management tool by downplaying negative information (Godfrey et al., 2009).

Consequently, when previously suppressed or overlooked negative information is eventually revealed, its market impact is often amplified. Prior studies further demonstrate that investors' emotional reactions to negative news are significantly stronger than their responses to positive information (Karampatsas et al., 2023). Media coverage of adverse corporate events frequently triggers immediate stock price declines (Akhtar et al., 2011). Some studies even find

that negative events lead to substantial reductions in enterprise market value, whereas positive events do not generate equivalent gains (Capelle-Blancard & Petit, 2019). In addition, negative information is often associated with market overreaction, while positive information tends to be underreacted to (Veronesi, 1999). Overall, these findings imply that capital markets are disproportionately responsive to negative corporate information relative to positive disclosures. Consequently, further analysis of the risk implications of negative ESG-related information is warranted.

From a signaling-theory viewpoint, sustainability reporting can be interpreted as an intentional message about what an enterprise expects to deliver in the future. Such disclosures may function as “forward-looking” cues regarding prospective performance (Lys et al., 2015) or forthcoming organizational adjustments (Lozano et al., 2015). The core premise of signaling theory is that, when uncertainty is high, credible signals help narrow the information gap between insiders and outsiders, allowing stakeholders to make better-informed judgments (Spence, 1974, 2002). Under this logic, sustainability disclosure goes beyond routine compliance or neutral documentation. Instead, it becomes a purposeful communication tool through which enterprises reveal otherwise unobservable qualities, such as internal resources, managerial competence, and organizational capabilities, to external audiences (Connelly et al., 2011).

Signaling theory further assumes rational behavior on the part of market actors. Enterprises therefore do not send signals randomly; they select and present information in ways that can influence how outsiders evaluate their credibility, competence, and reputation. In this setting, the adoption and disclosure of ESG practices may be read as evidence of an enterprise’s commitment to sustainability and social responsibility, as well as an indicator of professionalism in stakeholder and investor relations (Borgers et al., 2013; Liang & Cao, 2024; Orlitzky, 2013; Wang, 2024). On the receiving side, stakeholders who lack full access to internal information respond by actively seeking signals that are especially diagnostic of

enterprise-specific attributes, using them to infer expected performance and exposure to risk (Busenitz et al., 2005).

2.3.1 Theoretical Background

In markets where outsiders cannot fully observe an enterprise's true condition, decision-making is shaped by what can be credibly communicated. Signaling theory argues that well-designed signals help narrow the informational gap between insiders and external evaluators, allowing judgments to be formed under uncertainty with greater confidence (Spence, 1974, 2002). Sustainability communication fits naturally into this framework. Rather than functioning as a neutral record of past actions, sustainability-related disclosure can operate as a deliberate message about what the enterprise is likely to do next and how it is positioned for future outcomes. In this sense, it can be interpreted as a forward-oriented cue about expected performance (Lys et al., 2015) and, in some cases, as an indication of pending organizational adjustments (Lozano et al., 2015).

Viewed through this lens, disclosure becomes strategic. Enterprises use sustainability reporting to reveal qualities that are otherwise difficult to verify from the outside, such as internal resources, managerial competence, and governance capacity (Connelly et al., 2011). The theory also presumes purposive behavior by market actors: enterprises select signals with the intention of influencing how they are assessed, particularly with respect to credibility and reputation. ESG adoption and related disclosure can therefore be read as evidence that an enterprise is committed to sustainability objectives, takes social responsibilities seriously, and maintains disciplined relationships with investors and other stakeholders (Borgers et al., 2013; Liang & Cao, 2024; Orlitzky, 2013; Wang, 2024). Meanwhile, stakeholders who face limited access to private information do not remain passive. They actively look for disclosures that are especially diagnostic of enterprise-specific attributes and then use those cues to infer future performance prospects and exposure to risk (Busenitz et al., 2005).

From the perspective of negative signals, extensions of signaling theory suggest that signal senders may face a disadvantaged position in information transmission, thereby highlighting the critical role of feedback mechanisms in mitigating information disadvantages (Connelly et al., 2011). For enterprises, external feedback constitutes an important informational resource that helps assess whether the signals they emit achieve their intended effects. Such feedback can be broadly categorized into “hard feedback,” which is easily quantifiable, such as stock price movements and financial performance, and “soft feedback,” which is more difficult to measure directly, including corporate reputation and external criticism (Gamache & McNamara, 2019). Compared with hard feedback, soft feedback often exerts more persistent and far-reaching effects, as it reflects the collective evaluations of a broader range of stakeholders. These evaluations may impose sustained reputational pressure on enterprises and influence stakeholders’ perceptions over extended periods (Gamache & McNamara, 2019; Jung et al., 2018; Taghian et al., 2015). Among various sources of soft feedback, media coverage plays a particularly prominent role. The media not only serves as one of the most common channels through which negative signals are disseminated, but also represents a primary source through which enterprises gather feedback from stakeholders (Gamache & McNamara, 2019; Jung et al., 2018; Kölbel et al., 2017). Through media-generated negative feedback, enterprises are able to more directly perceive external evaluations of their actions and better understand stakeholders’ expectations.

Due to their immediacy and extensive information diffusion capacity, particularly in the digital environment, news media play a crucial role in facilitating information flows between enterprises and stakeholders. Media outlets enable information to spread rapidly and widely, thereby shaping stakeholders’ perceptions in a timely manner. At the same time, the media also serve as one of the primary channels through which enterprises can collect external stakeholders’ opinions and feedback (Argyris & Monu, 2015; Gamache & McNamara, 2019; Kölbel et al., 2017; Panico et al. 2014; Vogler et al., 2016). As noted by Gao et al. (2020), certain media

organizations actively verify, cross-check, and integrate information from multiple sources, transforming dispersed signals into economically meaningful information. This process helps reduce information frictions and enhances investors' ability to assess enterprise value more accurately. Through media coverage, enterprises' information environments become more transparent, which in turn strengthens investor trust and market participation. Moreover, empirical evidence suggests that media-provided information complements the formal disclosures contained in corporate annual reports and financial statements, thereby serving as an important reference for investors when evaluating enterprise conditions (Tetlock et al., 2008).

As one of the most influential sources of information in society, the media possess substantial power in attracting public attention and shaping social perceptions (Gamache & McNamara, 2019; Hussain & Ahmed, 2024; Kölbel et al., 2017; Okechukwu, 2023; Petkova, 2014). Consequently, enterprises can hardly ignore the potential adverse effects of negative media coverage on their reputation and risk assessment. Owing to the media's extensive dissemination capacity and credibility, negative media reports are likely to generate broadly unfavorable perceptions of enterprises among stakeholders (Breitinger & Bonardi, 2019; Fu, 2023; Gamache & McNamara, 2019; Peci, 2021; Voinea & Van Kranenburg, 2017). More importantly, such unfavorable media narratives may prompt stakeholders to take concrete actions, including initiating boycotts, terminating collaborations, withdrawing support, reducing or divesting investments, and even triggering higher financial risk exposure (Breitinger & Bonardi, 2019; Fu et al., 2021). In this sense, negative media coverage not only reflects societal evaluations but can also be directly transformed into tangible risks borne by enterprises.

Overall, enterprises voluntarily disclose positive ESG information and media-reported ESG-related negative events to send off two distinct sources of signals. Stakeholders integrate these signals when forming their overall assessments of enterprise risk. Negative media coverage thus serves as a critical feedback mechanism for evaluating enterprises' sustainability

practices and may exert a stronger influence on external market risk perceptions than enterprises' self-disclosed positive ESG information.

2.3.2 Hypothesis Development

Using the ESG framework as the organizing logic, this dissertation formulates hypotheses on how the environmental (E), social (S), and governance (G) pillars relate to enterprise risk. Beyond “positive” sustainability performance, the analysis also incorporates ESG-related negative event disclosures to reflect how adverse signals may alter the enterprise's risk profile when unfavorable information enters the public domain. By integrating sustainability performance indicators with negative ESG-related signals, the dissertation seeks to describe enterprise risk dynamics in a more complete and realistic manner.

Risk is typically separated into systematic risk, which cannot be eliminated through portfolio diversification, and idiosyncratic risk, which is more closely tied to enterprise-level choices and can be influenced by governance and managerial actions. Because the present study is concerned with risks that are responsive to internal strategies and governance design, it adopts idiosyncratic risk as the primary dependent variable.

2.3.2.1 Environmental Dimension

Recent ESG agendas increasingly prioritize climate-related and environmental risks, and this shift has begun to reshape corporate operating policies and governance practices. Prior evidence indicates that environmental exposure can raise enterprise risk and generate additional external monitoring costs. For instance, Truong et al. (2020) report that environmental risk is associated with higher overall risk and higher audit fees. At the supply-chain level, extreme climate events may disrupt production and logistics (e.g., food output and the movement of resources), amplifying operational uncertainty and fragility in global supply networks (Ghadge et al., 2020). When enterprises do not manage critical environmental issues, such as carbon

emissions, excessive water use, or severe pollution, they may face capital-market penalties, including divestment pressures or more restricted access to external financing.

A central mechanism emphasized in the environmental literature is the reputational and informational value of water-related transparency. Disclosing water information can help enterprises build reputational capital, strengthen stakeholder confidence, and reduce uncertainty and information asymmetry (Sun et al., 2025; Zeng et al., 2020). Reputational capital has long been viewed as a strategic asset that supports risk management and competitive advantage (Fombrun & Shanley, 1990). Because enterprise-specific risk is, in principle, diversifiable, one pathway to lowering idiosyncratic risk is to strengthen reputation through more proactive and credible water-related disclosure (Hasseldine et al., 2005). In addition, once an enterprise reaches a certain efficiency threshold, improvements in water-use efficiency may simultaneously raise profitability and reduce risk exposure (Fu and Jacobs, 2022; Gleick et al., 2011). Conversely, water constraints can increase production costs, damage reputation, and even threaten operating licenses (Kurland & Zell, 2010). Taken together, responsible water management and transparent reporting can reduce these exposures, and empirical studies find that environmental disclosure, especially water disclosure, is associated with lower idiosyncratic risk (Zhou et al., 2018).

Water disclosure can also be interpreted through a legitimacy channel. High-quality water disclosure may signal stronger organizational legitimacy, inviting broader stakeholder acceptance and support (Zeng et al., 2020). As peers imitate such practices, water disclosure may gradually become normalized within an industry. When institutionalized as a market expectation, disclosure can improve market stability by easing resource access, supporting enterprise value, lowering financing costs, and reducing uncertainty (Certo & Hodge, 2007; Delgado-García et al., 2013; Qiu et al., 2016). Building on these arguments, this dissertation expects that enterprises with higher water-use intensity face greater exposure to environmental

and operational uncertainty; as a result, higher water intensity should be associated with higher enterprise risk.

H1-A: Water intensity is positively related to idiosyncratic risk.

A parallel logic applies to carbon-related environmental exposures. Moral capital theory suggests that stronger non-financial performance, particularly on social and environmental issues, can function as an intangible asset that enhances stakeholder evaluations (Lundgren & Olsson, 2009). Empirical work further links greenhouse-gas management to economic outcomes: studies report that higher greenhouse gas emissions (or higher emission intensity) are associated with poorer financial performance (Galama & Scholtens, 2021; García-Sánchez & Prado-Lorenzo, 2012). More broadly, weak sustainability performance has been connected to greater exposure to both systematic and enterprise-specific risk (Boutin-Dufresne & Savaria, 2004; Lee & Faff, 2009; Lin et al., 2020; Luo & Bhattacharya, 2009). Under emissions-trading regimes, enterprise performance may also respond to regulatory settings. Evidence from the EU Emissions Trading System suggests that enterprises holding more allowances can exhibit stronger performance (Oestreich & Tsiakas, 2015). Related cross-country evidence using Kyoto-Protocol-related shocks indicates that high-emission enterprises may adjust capital structures (e.g., reduce leverage) when regulatory costs rise, underscoring the balance-sheet implications of carbon regulation for carbon-intensive enterprises (Nguyen & Phan, 2020).

Overall, the environmental strategy literature often finds that more proactive environmental approaches are associated with lower risk. In the context of greenhouse gas emissions, Konar and Cohen (2001) show that emissions reductions can translate into measurable enterprise value gains. Evidence from multiple regions, including China, the EU, and Japan, also suggests that higher emission levels are tied to higher carbon-related risk and weaker market outcomes (Jin et al., 2020; Oestreich & Tsiakas, 2015; Saka & Oshika, 2014).

Synthesizing these findings, this dissertation argues that greater carbon emission intensity should increase enterprise risk.

H1-B: Total carbon emission intensity is positively related to idiosyncratic risk.

H1-C: Scope 1 carbon emission intensity is positively related to idiosyncratic risk.

H1-D: Scope 2 carbon emission intensity is positively related to idiosyncratic risk.

2.3.2.2 Social Dimension

Research focusing on the social dimension of ESG consistently documents a negative association between corporate social responsibility and enterprise-level risk. Early evidence provided by Orlitzky and Benjamin (2001) indicates that CSR engagement is more strongly related to reductions in market-based risk indicators than to accounting-based risk measures. Subsequent studies reinforce this view. For instance, Dhaliwal et al. (2011) report that higher-quality CSR disclosure is associated with lower overall risk exposure. Similar patterns emerge in analyses that control for enterprise-specific characteristics, where superior CSR performance corresponds to reduced financial risk (Cai et al., 2016; Hsu & Chen, 2015).

Using a large-scale longitudinal dataset spanning 1995 to 2009, Kim et al. (2014) further show that enterprises with stronger CSR profiles exhibit greater transparency and a lower likelihood of stock price crash events. From the perspective of financing conditions and market access, CSR disclosure can also enhance enterprises' ability to attract customers and investors, thereby alleviating enterprise-specific risk (Li et al., 2014). Complementary evidence suggests that enterprises with more extensive CSR disclosure tend to maintain lower cash holdings, consistent with reduced precautionary motives under lower perceived risk (Cheung, 2016). In contrast, weak CSR performance has long been associated with elevated financial risk (McGuire et al., 1988). Taken together, this body of research suggests that CSR operates as a

substantive risk-mitigating mechanism rather than merely serving as a short-term reputational strategy.

Within CSR, employees are frequently treated as a central stakeholder group (Preuss et al., 2009). Beyond ethical considerations, attention to employee welfare can be strategically important because a stable, healthy, and engaged workforce supports long-run sustainability (González-Cantón et al., 2019). Workplace safety is a core element of this responsibility: safe working conditions, appropriate job design, and hazard reduction are foundational to protecting employee health (Benach et al., 2014). Employees also influence enterprise reputation and performance through attitudes, commitment, and organizational identification, which can affect product quality and customer loyalty and ultimately strengthen corporate image and competitiveness (Cravens & Oliver, 2006; Helm, 2011). Empirical evidence also links employee trust, rights protection, and related perceptions to stronger overall enterprise performance (Sakdanuwatwong, 2020; Sun & Yu, 2015).

When enterprises fail to manage occupational safety, such as neglecting hazard prevention or providing insufficient safety training, workplace exposure rises and injury rates may increase. Higher injury rates can generate medical, compensation, and legal costs, while also harming productivity, product quality, and customer satisfaction. These effects can erode reputation and profitability, thereby increasing risk. Accordingly, this dissertation argues that higher workplace injury rates are expected to be associated with higher enterprise risk.

H2-A: Workplace injury rates are positively related to idiosyncratic risk.

2.3.2.3 Corporate Governance Dimension

Governance mechanisms form a core part of internal control and decision systems and are closely tied to the interests of shareholders and other stakeholders (Babatunde & Olaniran, 2009;

Carney et al., 2011). Enterprises with stronger governance are generally better positioned to balance expected returns against risk exposure and to make more disciplined choices consistent with sound risk management. In the governance literature, two broad families of mechanisms are commonly analyzed: board characteristics and ownership structure (Abobakr & Elgiziry, 2016; Pergola & Joseph, 2011). These mechanisms influence how managerial behavior is monitored, how agency problems are constrained, and how decision quality is improved, thereby shaping risk outcomes (Filatotchev & Wright, 2011).

Boards influence corporate risk-taking by reviewing strategic proposals, providing guidance, and correcting managerial actions that deviate from long-term interests. Because strategic choices inherently involve risk (Amihud & Lev, 1981), boards must at the same time provide strategic oversight and constrain value-destroying managerial opportunism. Effective boards therefore need a broad monitoring perspective rather than narrow issue-by-issue responses (Bukhvalov & Bukhvalova, 2011). The degree to which boards can deliver this oversight depends on characteristics such as board size, independence (inside/outside composition), and gender diversity, which affect information access, deliberation quality, and monitoring strength.

Board monitoring is central because major corporate decisions typically require board scrutiny and approval. Board size, in particular, can proxy for the breadth of expertise and perspectives available in deliberations (Chaganti et al., 1985). Larger boards may also represent shareholder interests more fully and impose stronger monitoring pressure on management (Yermack, 1996). Prior studies frequently link larger board size to more stable outcomes and lower risk. For example, Cheng (2008) and Larmou and Vafeas (2010) associate larger boards with reduced performance volatility, and Pathan (2009) reports lower risk among large U.S. bank holding companies with larger boards. The common interpretation is that larger boards strengthen collective monitoring and encourage more cautious decision-making, thereby

reducing exposure to high-risk strategies (Cheng, 2008; Pathan, 2009). On this basis, this dissertation predicts a negative association between board size and idiosyncratic risk.

H3-A: Board size is negatively related to idiosyncratic risk.

Board independence is widely viewed as key governance device that can restrain opportunistic behavior and excessive managerial risk-taking. Independent directors are often characterized as playing a monitoring role that is more objective than that of inside directors (Davis & Kay, 1990; Zahra, 1993). Empirical work also finds that higher independence can improve oversight effectiveness (Albring et al., 2014). Because independent directors are not involved in daily management (Buchholtz & Ribbens, 1994) and their status depends heavily on reputation, they may be particularly sensitive to the consequences of risky corporate actions and potential liability (Zajac & Westphal, 1996). Evidence also suggests reputational penalties for directors of distressed enterprises (Gilson, 1990). From a resource dependence perspective, independent directors can also help enterprises access external information and resources and enhance legitimacy (Peng, 2004). Additional arguments suggest they can constrain value-destroying opportunism and strengthen governance professionalism, which supports minority shareholder protection (Peng, 2004).

As governance becomes more stringent, managerial discretion tends to narrow, encouraging more prudent decisions aligned with multiple stakeholder interests and potentially reducing earnings volatility and overall risk (Adams et al., 2005). The expertise of independent directors can further matter: Liu and Sun (2024) find that a higher share of professionally qualified independent directors is associated with lower total and systemic risk. Taken together, stronger board independence is expected to improve monitoring and reduce operational and compliance risks, leading to lower idiosyncratic risk.

H3-B: The proportion of independent directors is negatively related to idiosyncratic risk.

Gender diversity on boards is often linked to differences in risk preferences and ethical orientations. Prior work suggests female directors are, on average, more risk-averse and less tolerant of opportunistic behavior (Carter et al., 2017; Cumming et al., 2015). Female directors can strengthen communication and collaborative governance climates, thereby transmitting more cautious values to management and constraining inappropriate behavior. Evidence indicates women directors may show stronger monitoring incentives and more prudent governance-related decisions, supporting governance quality (Adams & Ferreira, 2009; Srinidhi et al., 2020). Similar patterns appear in executive behavior: female executives often favor less risky investment and financing choices, and enterprises led by women may exhibit lower risk exposure (Eckel & Füllbrunn, 2015; Faccio et al., 2016). Prior findings suggest such risk aversion can shape corporate strategies toward more conservative choices and lower overall risk (Faccio et al., 2016). Accordingly, this dissertation expects that a higher proportion of female directors is associated with lower idiosyncratic risk.

H3-C: The proportion of female directors on the board is negatively related to idiosyncratic risk.

Ownership structure is a classic mechanism for addressing agency problems and reducing agency costs (Meckling & Jensen, 1976). Theoretical and empirical research has long argued that ownership allocation affects governance effectiveness and enterprise behavior; it shapes organizational design, monitoring intensity, and managerial incentives, which in turn influence operating outcomes (Connelly et al., 2010). In line with this view, this dissertation examines ownership structure through two primary lenses: the shareholding ratio of large shareholders and managerial ownership.

An argument proposes that large shareholders strengthen governance because they have both the incentive and the capacity to monitor. With substantial economic exposure and meaningful voting power, large shareholders can discipline managers and intervene when actions threaten shareholder value (Jensen, 1993; Shleifer & Vishny, 1997). Because they internalize more of the benefits from improved governance, they may be more willing to pay monitoring costs. Consistent with an “active monitoring” logic, evidence links higher large shareholder ownership to more intense oversight, fewer agency problems, and stronger involvement in enterprise decisions (Brickley et al., 1988; Demsetz, 1983; McConnell & Servaes, 1995; Shleifer & Vishny, 1986). Some studies further argue that multiple large shareholders may reduce dominance by a single controlling owner, temper entrenchment, and provide additional checks that protect minority shareholders (Mishra, 2011). Empirical findings are often interpreted as supporting more efficient monitoring and potentially lower operating risk when large shareholders are present (Brickley et al., 1988; McConnell & Servaes, 1990; Morck et al., 1988; Shleifer & Vishny, 1997).

However, an alternative perspective highlights potential costs of concentrated ownership. Under wealth-transfer arguments, highly concentrated control may allow large shareholders to extract private benefits or pursue strategies that increase risk (Barclay & Holderness, 1989). Such behavior can include high-risk investment, higher leverage, or resource transfers through dividends or related-party transactions, potentially harming minority shareholders and other stakeholders (Liu et al., 2022). In these settings, control can elevate both operating and financial risks, particularly if governance is used to pursue private interests rather than long-term value creation (Morck et al., 2005; Shleifer & Vishny, 1997). In addition, Bethel and Liebeskind (1993) argue that large shareholders may prevent managers or other insiders from adopting risk-reducing corporate strategies, as such strategies may, in certain circumstances, conflict with the large shareholder's own interests. As a result, large shareholder ownership can reduce risk via monitoring or increase risk via expropriation. Based on the aforementioned mechanisms,

this dissertation expects that a higher level of large shareholder ownership is associated with an increase in idiosyncratic risk.

H3-D: The ownership share of large shareholders is positively related to idiosyncratic risk.

According to agency theory, agency costs stem from the combination of information asymmetry and misaligned incentives between owners and managers (Meckling & Jensen, 1976). Because ownership is separated from control, managers may pursue objectives that do not fully coincide with shareholders' wealth maximization, creating room for opportunistic behavior. One commonly discussed solution is to increase managers' equity exposure. When managers hold more shares, their personal wealth becomes more sensitive to enterprise outcomes, which can strengthen their motivation to select actions that enhance enterprise value (Meckling & Jensen, 1976). Empirical findings are consistent with this alignment argument. Mehran (1995) shows that managerial ownership can ease conflicts between shareholders and top executives and is positively associated with enterprise performance. The informational advantage of managers in daily operations further reinforces the relevance of equity incentives: because managers often have faster and more complete internal information than owners, they may otherwise exploit that advantage for private benefit. Equity-based compensation and ownership stakes are therefore widely treated as mechanisms that reduce agency frictions by aligning managerial and shareholder interests. Supporting this view, Zhang et al. (2021) argue that when managers hold residual claims, their decisions are more likely to reflect long-term enterprise interests, which can improve managerial efficiency and foster steadier development. Evidence discussed by Morck et al. (1988) similarly emphasizes that managerial ownership may lower agency costs and strengthen governance effectiveness.

In addition, ownership concentration among managers can also influence risk through stronger monitoring. When managers hold sizable stakes, they possess both the capacity and

the incentive to supervise management more actively, because their wealth is closely tied to enterprise performance. More intensive oversight may improve decision discipline, strengthen governance quality, and reduce risk exposure. Following this logic, prior research often implies that greater monitoring pressure associated with higher managerial ownership can contribute to lower enterprise risk. Building on these arguments, this study expects managerial ownership to be related to lower idiosyncratic risk.

H3-E: Managerial ownership is negatively related to idiosyncratic risk.

2.3.2.4 ESG-Related Negative Events

Much of the ESG literature has concentrated on favorable ESG signals, asking whether strong ESG performance lowers risk and improves financing conditions. Positive ESG-related media tone, for example, may be interpreted as suggesting lower environmental exposure, stronger stakeholder support, and a reduced likelihood of opportunistic reporting behavior. Such favorable impressions can strengthen market confidence in the enterprise's offerings and improve long-horizon expectations. In credit markets, these reputational cues can help lenders identify borrowers perceived as more stable and less prone to default, thereby reducing adverse-selection concerns before lending occurs. Media attention may also play an external monitoring role by drawing public and regulatory scrutiny, which can reduce banks' monitoring burdens after lending and increase willingness to provide longer-term credit (Chen et al., 2021; Yang et al., 2014). From a stakeholder theory lens, positive ESG recognition can be read as a form of social legitimacy on issues valued by key stakeholder groups, lowering informational frictions and transaction costs in relationships with banks, investors, and suppliers. Consistent with this view, prior studies associate stronger ESG performance with more favorable borrowing terms and reduced financing costs (El Ghouli et al., 2021; Kashyap et al., 2025; Raimo et al., 2021).

Cheaper and more abundant financing can improve financial flexibility and support more efficient asset allocation and balance-sheet structure. Moreover, positive ESG reporting can add to reputational capital and lower risk premia in long-term lending by reducing information asymmetry (Zhang et al., 2024). Through these channels, positive ESG signals may lower both operational risk and market-related risk and strengthen overall financial stability (Rezaee & Tuo, 2019).

Negative ESG information, however, is often processed differently and can trigger disproportionately adverse market responses. Empirical studies suggest that negative disclosures tend to produce sharper reactions than comparably positive news (Bird et al., 2007; Sabbaghi, 2022). Bénabou and Tirole (2010) argue that enterprises with strong social-responsibility reputations can be viewed as embodying civic values, making them subject to higher expectations and more intense scrutiny from consumers, investors, NGOs, and other external audiences. Related work shows that setbacks in environmental performance can reduce returns while simultaneously increasing operating costs, implying an unfavorable double effect (Klassen & McLaughlin, 1996). Their evidence further indicates that abnormal returns associated with negative environmental news are typically larger in magnitude than gains associated with positive news, pointing to asymmetric valuation effects. At the enterprise level, Kim et al. (2014) suggest that stronger CSR performance can lower subsequent crash risk partly because such enterprises are less likely to stockpile undisclosed bad news; accordingly, when an enterprise with a responsible image is hit by an adverse event, the reputational loss and market punishment can be particularly severe.

Capital-market pricing of environmental performance provides further context for why negative ESG events can matter. Enterprises with lower greenhouse gas emissions have been found to face more favorable financing conditions, including lower costs of equity and debt, implying a lower overall cost of capital (Jung et al, 2018; Kim et al., 2015; Lemma et al., 2019). This pattern suggests that investors and creditors accept lower required returns for

environmentally responsible enterprises, reflecting more favorable perceptions. In contrast, heavy polluters may be penalized through higher required returns and weaker market evaluations. During volatile periods or large negative shocks, investors may become less tolerant of poor environmental performance and reallocate capital away more quickly, adding downward pressure to prices and increasing vulnerability to market and financial risk.

A behavioral channel strengthens this asymmetry: negativity bias. This bias describes the tendency for negative information to be processed more rapidly and to weigh more heavily than positive information of similar magnitude. Evidence from equity markets supports this claim. Akhtar et al. (2011) document a pronounced negativity effect in the Australian market, where adverse news exerts stronger downward pressure on stock prices. Reyes (2019) similarly finds that negative market movements attract greater attention from retail investors than comparable positive changes, indicating heightened sensitivity to unfavorable information. Psychological research also suggests that negative cues carry greater weight and persist longer in shaping overall impressions (Baumeister et al., 2001; Peeters & Czapinski, 1990; Sheldon et al., 1996). Taken together, these mechanisms imply that ESG-related negative events can amplify pessimistic beliefs about an enterprise and strengthen market assessments of risk. Therefore, this dissertation expects that a higher frequency of ESG-related negative events corresponds to higher idiosyncratic risk.

H4-A: The frequency of ESG-related negative events is positively related to idiosyncratic risk.

Stakeholders may also respond more intensely when negative ESG events are more severe, because severe events can impose larger reputational losses and more substantial valuation impacts (Kölbel et al., 2017; Palmrose et al., 2004). When adverse events involve larger penalties, more serious allegations, or repeated violations, the market response is likely to be stronger, reflecting greater expected damage to reputation and future cash flows. In other words,

severity can magnify the perceived implications for shareholder value and elevate perceived risk. To capture this idea empirically, this dissertation measures severity using the ratio of ESG-related fines to total assets. Under this proxy, larger fines relative to enterprise size indicate more severe negative events. Building on prior literature and the logic that severity intensifies reputational and financial consequences, this dissertation predicts that greater severity of ESG-related negative events will be associated with higher idiosyncratic risk.

H4-B: The severity of ESG-related negative events is positively related to idiosyncratic risk.



III. Dissertation Methodology

This dissertation adopts a two-stage empirical framework. In the first stage, idiosyncratic risk measures are estimated. In the second stage, the relationships between ESG indicators across different dimensions and idiosyncratic risk are empirically examined.

3.1 First Stage-Estimation of Idiosyncratic Risk

Idiosyncratic risk refers to the component of an enterprise's total risk that is not attributable to market-wide movements and persists after systematic risk has been accounted for (Sassen et al., 2016). Consistent with prevailing empirical practice, this dissertation derives idiosyncratic risk from the residual terms of the Fama and French (2015) five-factor asset pricing model. This specification is particularly appropriate because it incorporates a richer set of risk factors relevant to enterprise-level stock return variation, thereby offering a more encompassing representation of enterprise-specific risk exposure (Fama & French, 2015; Luo & Bhattacharya, 2009; Sassen et al., 2016). Comparative evidence further supports this choice. Zaremba and Czapkiewicz (2017) evaluate the explanatory performance of the Capital Asset Pricing Model (CAPM), the Fama–French three-factor model, and the five-factor model, and demonstrate that the five-factor specification outperforms alternative models in explaining portfolio returns. In addition, a growing body of empirical research documents the robustness and reliability of the five-factor model across diverse samples and empirical contexts (Chiah et al., 2016; Huynh, 2018). The model is:

$$R_{it} - R_{ft} = \alpha_i + \beta_1 MR_t + \beta_2 SMB_t + \beta_3 HML_t + \beta_4 CP + \beta_5 Asset_G + e_{it} \quad (1)$$

where R_{it} denotes the return of enterprise i in period t . R_{ft} represents the risk-free rate in period t . $R_{it} - R_{ft}$ (ER) measures the excess return of enterprise i at time t . The market factor

(MR) is constructed as the return on the Taiwan Stock Exchange portfolio net of the risk-free rate, with the latter proxied by the yield on three-month, non-listed government bonds. The size factor (SMB) reflects the return spread between portfolios composed of small-cap enterprises and those consisting of large-cap enterprises. The value factor (HML) measures the return differential between enterprises with high versus low book-to-market ratios. Enterprise profitability is captured by the profitability factor (CP), calculated as operating income scaled by book value and used to represent operating performance. The investment factor is proxied by recent asset growth, where the asset growth rate (Asset_G) is defined as the change in total assets over the preceding six-month period. Finally, e_{it} represents the regression residual, capturing the portion of enterprise returns that cannot be explained by the five factors and is used as a measure of idiosyncratic risk. The regression residuals are subsequently standardized to construct the proxy for idiosyncratic risk.

3.2 Second Stage-Empirical Model Estimation

In the empirical analysis, idiosyncratic risk serves as the dependent variable. The explanatory variables are organized into four broad categories: environmental performance indicators, social performance indicators, corporate governance characteristics, and ESG-related adverse event measures.

Environmental performance is captured through multiple intensity-based indicators, including water intensity, aggregate carbon emission intensity, and disaggregated Scope 1 and Scope 2 carbon emission intensities. Water intensity is employed as a proxy for environmental efficiency and is calculated as total water consumption normalized by operating revenue in year 2021 (Liu, 2023; Tian & Adriaens, 2025). Carbon emission intensity is likewise a widely adopted measure of enterprises' environmental footprint (Drempetic et al., 2020; Hoffmann & Busch, 2008; Thomakos & Alexopoulos, 2016; Wedari et al., 2023). Consistent with Kleimeier and Viehs (2018), Scope 1 and Scope 2 emissions are examined separately to differentiate

between direct and indirect sources of carbon emissions, with each intensity measure defined as the corresponding emissions scaled by revenue in year 2021.

Social performance is proxied by the workplace injury rate, defined as the number of recordable lost-time injuries relative to total hours worked. This indicator is commonly used in ESG disclosure frameworks to assess occupational safety and employee well-being (Hallowell et al., 2024; Yoon et al., 2025), and it reflects enterprises' internal social responsibility toward their workforce.

Corporate governance variables are operationalized along two dimensions: board structure and ownership configuration. Board-related characteristics include board size, the proportion of independent directors, and the proportion of female directors, which prior studies associate with monitoring effectiveness and decision quality (Duchin et al., 2010; Horváth & Spirollari, 2012; Reguera-Alvarado & Bravo, 2017; Shukeri et al., 2012). Ownership structure is captured through the equity holdings of large shareholders and managerial ownership, representing ownership concentration and incentive alignment, respectively (Courteau et al., 2017; King & Wen, 2011; Sarhan & Al-Najjar, 2023).

To account for ESG-related negative signals, the study incorporates adverse-event information following de Vincentiis (2023). Major ESG-related incidents documented in the TEJ database are identified, and each enterprise's annual frequency of such negative events is calculated as an indicator of negative ESG exposure. In addition, event severity is proxied by environmental penalty fines scaled by total assets, capturing the magnitude of regulatory sanctions associated with ESG-related violations.

Consistent with prior empirical research (Bouslah et al., 2013; Chairani & Siregar, 2021), the regression specifications further control for six enterprise-level characteristics: enterprise size (ESIZE), profitability (ROA), financial leverage (FLEVE), cash holdings (CASHR), asset growth (ASGR), and revenue growth (REGR). Enterprise size is measured as the natural logarithm of total assets in year 2021. Financial leverage is defined as total assets divided by

shareholders' equity. CASHR is calculated as cash and cash equivalents relative to total assets. Asset growth (ASGR) and revenue growth (REGR) are measured as period-to-period changes in total assets and operating revenue, respectively, each scaled by their lagged values.

3.3 Data Sources

The data consists of enterprises listed on the Taiwan Stock Exchange during 2018–2024. The start year is set to 2018 because Taiwan launched a revised Corporate Governance Roadmap and introduced sustainability indices in that year, making it a meaningful baseline for ESG-related analysis. The end year is 2024 because it is the most recent year with complete financial and ESG data coverage for the study.

A total of 314 listed enterprises is identified for the period. Enterprises in financial and insurance industries are removed because their business models and regulatory constraints differ materially and may distort comparisons (Fuente et al., 2017). The final panel includes 314 enterprises over seven years, yielding 2,198 enterprise-year observations.

Table 1 summarizes the definitions and data sources of all variables employed in this study. All key variables—water consumption, total emissions, Scope 1 and Scope 2 emissions, injury rate, board size, independent-director proportion, female-director proportion, large-shareholder ownership, managerial ownership, negative-event frequency, and negative-event severity—are obtained from TEJ. Severity is computed as monetary penalties for ESG-related violations scaled by total assets, capturing the relative magnitude of regulatory sanctions faced by each enterprise.

Table 1. List of Variables

Variable	Name	Symbol	Definition	Unit	Source
Stage 1-Estimation of Idiosyncratic Risk					
	Excess return	ER	The excess return of enterprise i in period t, calculated as the enterprise's realized return net of the contemporaneous risk-free rate	%	TEJ
	Market factor	MR	The market factor, defined as the return on the Taiwan Stock Exchange portfolio minus the risk-free rate, with the latter proxied by the yield on three-month non-listed government bonds	%	TEJ
	Size factor	SMB	The size factor, capturing the return spread between portfolios composed of small-cap enterprises and those consisting of large-cap enterprises	%	TEJ
	Value factor	HML	The value factor, measured as the return differential between portfolios with high and low book-to-market (B/M) ratios.	%	TEJ
	Profitability factor	CP	The profitability factor, calculated as operating income scaled by book value, representing enterprises' operating profitability	%	TEJ
	Investment factor	Asset_G	The investment factor, proxied by recent asset growth, where the asset growth rate reflects changes in total assets over the preceding six-month period	%	TEJ
	Idiosyncratic risk	IDIRISK	The idiosyncratic risk measure, constructed from the standardized residuals obtained from the five-factor regression model	-	This study
Stage 2-Empirical Model Estimation					
Dependent	Idiosyncratic risk	IDIRISK	Idiosyncratic risk, measured using the standardized residuals obtained from the Fama-French five-factor regression model	-	This study
Independent	Water intensity	WI	Water intensity, calculated as total water consumption scaled by net operating revenue in year 2021	Metric tons / Thousand NTD	This study
	Total carbon intensity	TCI	Total CO2 emission divided by net operating revenue in year	Metric tons /	This study

			2021		Thousand NTD	
	Scope 1 carbon emission intensity	S1	Scope 1 CO2 emission divided by net operating revenue in year 2021	Metric tons / Thousand NTD	This study	
	Scope 2 carbon emission intensity	S2	Scope 2 CO2 emission divided by net operating revenue in year 2021	Metric tons / Thousand NTD	This study	
	Injury rate	WIR	Occupational injury rate, measured as the number of lost-time injuries per 200,000 working hours	-	TEJ	
	Board size	BS	Total number of directors on the board	Count	TEJ	
	Proportion of independent directors	IDR	Ratio of independent directors to total board members	-	This study	
	Proportion of female directors	FDR	Ratio of female directors to total board members	-	This study	
	Ownership concentration of large shareholders	MSR	Shareholding ratio of large shareholders disclosed in monthly public information and annual shareholders' meetings	%	TEJ	
	Managerial ownership	MAR	Shareholding ratio of managers, including directors and senior executives	%	TEJ	
	Frequency of ESG-related negative events	NEF	Total number of ESG-related negative events reported for the enterprise during the year	Count	TEJ (Mass Media)	
	Severity of ESG-related negative events	NEI	Total penalty amount divided by total assets	-	This study	
Control	Enterprise size	ESIZE	Enterprise size, measured as the natural logarithm of total assets in year 2021	-	This study	
	Return on assets	ROA	Return on assets (ROA), defined as net income scaled by total assets	-	TEJ	
	Financial leverage	FLEVE	Financial leverage, calculated as total liabilities divided by total assets	-	This study	
	Cash asset ratio	CASHR	Cash and cash equivalents divided by total assets	-	This study	
	Asset growth rate	ASGR	Annual growth rate of total assets	-	This study	
	Revenue growth rate	REGR	Annual growth rate of operating revenue	-	This study	

3.4 Research Method

3.4.1 Methodology

A further concern in ESG and governance research is endogeneity, including omitted variables and reverse causality, which may bias estimates (Ahmad et al., 2025; Goergen et al., 2019; Wintoki et al., 2012). To mitigate these issues and strengthen causal interpretation, the study additionally adopts two-stage least squares (2SLS) estimation (Maydeu-Olivares et al., 2019; Osei, 2025; Saikia, & Maji, 2026; Ullah et al., 2021). Effective 2SLS requires instruments that satisfy both relevance and exogeneity (Ullah et al., 2021).

For water intensity and carbon emission intensity, the study follows the literature by using industry-average intensity measures as instruments. Industry-level metrics have been used as plausible instruments for enterprise-level resource-use and emission outcomes because they correlate strongly with enterprise practices while being less directly determined by any single enterprise (Li & Tian, 2023; Nyiwul et al., 2025; Sun et al., 2025). Accordingly, average water intensity and carbon intensity computed at the 27-industry level are adopted as IVs. The logic is: (1) enterprise environmental intensity tends to co-move with industry resource-use and emission patterns (relevance), and (2) industry averages are less likely to be driven by enterprise-specific shocks to idiosyncratic risk (supporting exogeneity).

In addition, following the approach of Chen et al. (2025), this dissertation adopts the one-period lag of the work injury rate as an instrumental variable, given its strong correlation with the current workplace injury rate and its relatively weak association with the contemporaneous error term, which helps mitigate endogeneity concerns (Bai et al., 2019; Dai, 2025; Han et al., 2025). For the corporate governance dimension, this dissertation employs industry-average board characteristics variables and the lagged values of ownership structure variables as instrumental variables to capture dynamic adjustments in governance structures over time (Bhagat & Bolton, 2008; Goergen et al., 2019; Islam et al., 2025; Mbanyele & Wang, 2022;

Ntim et al., 2015; Pan et al., 2018). Industry-level board characteristics variables reflect differences in governance environment and resource allocation; however, its influence on short-term price fluctuations and market liquidity is limited, implying a weaker direct impact on enterprise risk, making it a suitable exogenous instrument (Hsu et al., 2020; Li et al., 2020).

Furthermore, although governance indicators capture current ownership structure, adjustments in governance structures generally occur with time lags and are not immediately linked to contemporaneous risk shocks. Therefore, using lagged governance variables as instruments is theoretically reasonable, as past governance structures may influence current sustainability or governance decisions, while remaining less correlated with present error terms (Osei, 2025; Talalwa et al., 2025). When the relevance of a one-period lag is insufficient, second-lag or multiple-lag instruments can be employed to improve identification strength and enhance the robustness of estimation results (Wintoki et al., 2012).

Finally, regarding the instrumental variable selection for ESG-related negative event indicators, this dissertation follows Harjoto and Laksmana (2018) and Luo et al. (2024) by employing industry-average ESG negative events indicators as instrumental variables. Enterprises operating in the same industry typically display comparable disclosure behaviors, influenced by common regulatory frameworks, industry practices, and external oversight. As a result, the industry-level disclosure frequency is unlikely to directly affect an individual enterprise's contemporaneous risk or investment decisions, thereby satisfying the exclusion restriction (Dyck et al., 2019; Liao, 2025).

3.4.2 Structure the Models

This dissertation adopts the two-stage least squares (2SLS) estimation technique in light of prior evidence suggesting the presence of a reciprocal relationship between corporate social performance and enterprise-level risk (Bouslah et al., 2013). Specifically, ESG disclosure may affect an enterprise's risk exposure, while enterprises with different risk profiles may

simultaneously adjust their ESG disclosure strategies. This potential reverse causality implies that ESG-related variables are likely to be endogenous, which could bias coefficient estimates if not properly addressed. To alleviate these endogeneity concerns and to strengthen the validity of causal interpretation as well as the robustness of the empirical findings, the study therefore employs a 2SLS framework in the empirical analysis.

To assess the effects of the environmental, social, and governance dimensions on idiosyncratic risk, and to further evaluate how the frequency and severity of ESG-related adverse events influence idiosyncratic risk, the dissertation specifies a series of 2SLS regression models. These models are designed to examine the relationships between the explanatory variables and idiosyncratic risk. The formal specifications of the 2SLS models are presented below:

3.4.2.1 Environmental Dimension:

Stage 1:

$$WI_{it} = \gamma_i + \theta_1 \text{Industry_}WI_{it} + \theta_2 \text{CONTROL}_{it} + \delta_{1it} \quad (1a_1)$$

$$TC_{it} = \gamma_i + \theta_1 \text{Industry_}TC_{it} + \theta_2 \text{CONTROL}_{it} + \delta_{2it} \quad (1b_1)$$

$$S1_{it} = \gamma_i + \theta_1 \text{Industry_}S1_{it} + \theta_2 \text{CONTROL}_{it} + \delta_{3it} \quad (1c_1)$$

$$S2_{it} = \gamma_i + \theta_1 \text{Industry_}S2_{it} + \theta_2 \text{CONTROL}_{it} + \delta_{4it} \quad (1d_1)$$

Stage 2:

$$\text{IDIRISK}_{it} = \omega_i + \vartheta_1 \widehat{WI}_{it} + \vartheta_2 \text{CONTROL}_{it} + \zeta_{1it} \quad (1a_2)$$

$$\text{IDIRISK}_{it} = \omega_i + \vartheta_1 \widehat{TC}_{it} + \vartheta_2 \text{CONTROL}_{it} + \zeta_{2it} \quad (1b_2)$$

$$\text{IDIRISK}_{it} = \omega_i + \vartheta_1 \widehat{S1}_{it} + \vartheta_2 \text{CONTROL}_{it} + \zeta_{3it} \quad (1c_2)$$

$$\text{IDIRISK}_{it} = \omega_i + \vartheta_1 \widehat{S2}_{it} + \vartheta_2 \text{CONTROL}_{it} + \zeta_{4it} \quad (1d_2)$$

3.4.2.2 Social Dimension:

Stage 1:

$$WIR_{it} = \gamma_i + \theta_1 WIR_{it-1} + \theta_2 CONTROL_{it} + \delta_{5it} \quad (2a_1)$$

Stage 2:

$$IDIRISK_{it} = \omega_i + \vartheta_1 \widehat{WIR}_{it} + \vartheta_2 CONTROL_{it} + \zeta_{5it} \quad (2a_2)$$

3.4.2.3 Corporate Governance Dimension:

Stage 1:

$$BS_{it} = \gamma_i + \theta_1 Industry_BS_{it} + \theta_2 CONTROL_{it} + \delta_{6it} \quad (3a_1)$$

$$IDR_{it} = \gamma_i + \theta_1 Industry_IDR_{it} + \theta_2 CONTROL_{it} + \delta_{7it} \quad (3b_1)$$

$$FDR_{it} = \gamma_i + \theta_1 Industry_FDR_{it} + \theta_2 CONTROL_{it} + \delta_{8it} \quad (3c_1)$$

$$MSR_{it} = \gamma_i + \theta_1 MSR_{it-1} + \theta_2 CONTROL_{it} + \delta_{9it} \quad (3d_1)$$

$$MAR_{it} = \gamma_i + \theta_1 MAR_{it-1} + \theta_2 CONTROL_{it} + \delta_{10it} \quad (3e_1)$$

Stage 2:

$$IDIRISK_{it} = \omega_i + \vartheta_1 \widehat{BS}_{it} + \vartheta_2 CONTROL_{it} + \zeta_{6it} \quad (3a_2)$$

$$IDIRISK_{it} = \omega_i + \vartheta_1 \widehat{IDR}_{it} + \vartheta_2 CONTROL_{it} + \zeta_{7it} \quad (3b_2)$$

$$IDIRISK_{it} = \omega_i + \vartheta_1 \widehat{FDR}_{it} + \vartheta_2 CONTROL_{it} + \zeta_{8it} \quad (3c_2)$$

$$IDIRISK_{it} = \omega_i + \vartheta_1 \widehat{MSR}_{it} + \vartheta_2 CONTROL_{it} + \zeta_{9it} \quad (3d_2)$$

$$IDIRISK_{it} = \omega_i + \vartheta_1 \widehat{MAR}_{it} + \vartheta_2 CONTROL_{it} + \zeta_{10it} \quad (3e_2)$$

3.4.2.4 ESG-Related Negative Events:

Stage 1:

$$NEF_{it} = \gamma_i + \theta_1 Industry_NEF_{it} + \theta_2 CONTROL_{it} + \delta_{11it} \quad (4a_1)$$

$$NEI_{it} = \gamma_i + \theta_1 Industry_NEI_{it} + \theta_2 CONTROL_{it} + \delta_{12it} \quad (4b_1)$$

Stage 2:

$$\text{IDIRISK}_{it} = \omega_i + \vartheta_1 \widehat{\text{NEF}}_{it} + \vartheta_2 \text{CONTROL}_{it} + \zeta_{11it} \quad (4a_2)$$

$$\text{IDIRISK}_{it} = \omega_i + \vartheta_1 \widehat{\text{NEI}}_{it} + \vartheta_2 \text{CONTROL}_{it} + \zeta_{12it} \quad (4b_2)$$

where subscripts i and t refer to enterprise and year, respectively. IDIRISK represents idiosyncratic risk. WI denotes water intensity, TC refers to total carbon emission intensity, while S1 and S2 capture Scope 1 and Scope 2 carbon emission intensities, respectively. WIR indicates the workplace injury rate. Board-related governance characteristics include BS, measured as board size, IDR, defined as the proportion of independent directors, and FDR, representing the proportion of female directors on the board. Ownership structure variables consist of MSR, which measures ownership concentration among large shareholders, and MAR, denoting managerial ownership. Finally, NEF captures the annual frequency of ESG-related adverse events. NEI denotes the severity of ESG-related negative events. Industry_WI_{it} , Industry_TC_{it} , Industry_S1_{it} , Industry_S2_{it} , Industry_BS_{it} , Industry_IDR_{it} , Industry_FDR_{it} , Industry_NEF_{it} , and Industry_NEI_{it} represent the industry-level averages of WI, TC, S1, S2, BS, IDR, FDR, NEF, and NEI of peer enterprises, respectively. WIR_{it-1} , MSR_{it-1} , and MAR_{it-1} denote the one-period lagged values of WIR, MSR, and MAR, respectively. $\widehat{\text{WI}}_{it}$, $\widehat{\text{TC}}_{it}$, $\widehat{\text{S1}}_{it}$, $\widehat{\text{S2}}_{it}$, $\widehat{\text{WIR}}_{it}$, $\widehat{\text{BS}}_{it}$, $\widehat{\text{IDR}}_{it}$, $\widehat{\text{FDR}}_{it}$, $\widehat{\text{MSR}}_{it}$, $\widehat{\text{MAR}}_{it}$, $\widehat{\text{NEF}}_{it}$, and $\widehat{\text{NEI}}_{it}$ represent the predicted values of WI, TC, S1, S2, WIR, BS, IDR, FDR, MSR, MAR, NEF, and NEI for enterprise i in year t , which are obtained from the first-stage regression estimation. CONTROL represents control variables, including ESIZE, ROA, FLEVE, CASHR, ASGR, and REGR. δ and ζ are the error terms, following a normal distribution $N(0, \sigma^2)$, indicating a mean of zero and a constant variance.

IV. Empirical Results and Discussion

4.1 Data Distribution

According to the data distribution presented in Table 2, this dissertation includes 314 enterprises across 27 industry categories. The data are primarily composed of high-tech and manufacturing industries, among which semiconductor (10.51%), chemical industry (10.19%), electronic components (8.60%), and computer and peripheral equipment (8.60%) represent the four largest sectors, accounting for approximately 37.9% of the total observations. This indicates that the dataset is characterized by a high proportion of capital-intensive and technology-oriented industries. Overall, the dataset is mainly sourced from electronics-related sectors, such as semiconductors, electronic components, and computer peripherals, whose distribution pattern generally aligns with that of Taiwan's listed enterprises, suggesting that the dataset holds a certain degree of market representativeness.

4.2 Descriptive Statistics and Correlation Analysis

Table 3 presents the descriptive statistics of the observations used in this dissertation. First, the environmental variables exhibit clear patterns. The average total carbon emission intensity is 4.16, with a median of 3.25. Scope 1 (direct emissions) carbon intensity has an average of 4.01 and a median of 3.08, while Scope 2 (indirect emissions) carbon intensity shows an average of 4.20 and a median of 3.25. These results suggest that carbon emission intensities across different emission sources exhibit similar distributional patterns, indicating that enterprises' overall carbon emission structures are characterized by a certain degree of diversity and complexity.

Regarding ESG negative event indicators, the average frequency of negative events is 4.27, with a median of 1. This implies that most enterprises experienced at least one ESG-related negative event during the data period, while a subset of enterprises accumulated a relatively high number of such events. The severity of ESG-related negative events, measured as the ratio

of penalty amounts to total assets, has an average of 0.02 and a median of 0. This indicates that although ESG-related negative events are not uncommon, the overall monetary penalties relative to enterprise asset size are generally small, and their immediate financial impact is limited. Overall, the enterprises in the dataset exhibit a non-negligible frequency of ESG-related negative events. However, the severity of these events, in terms of direct financial penalties, appears to be relatively modest.

This dissertation conducts multicollinearity diagnostics using the Pearson correlation matrix (Table 4) and variance inflation factors (VIFs) (Tables 5) to examine whether strong linear relationships exist among the independent variables. The pairwise correlation matrix indicates that most correlations are relatively low, with only a few variable pairs exhibiting high correlations, such as a correlation coefficient of 0.99 between TC and S1, 0.95 between TC and S2, and 0.97 between S1 and S2. Nevertheless, the VIF results reported in Tables 5 show that all variables have VIF values well below the commonly accepted threshold of 10 (Alin, 2010), suggesting that multicollinearity does not pose a serious concern in this dissertation. Overall, the results indicate a low risk of multicollinearity, supporting the robustness and reliability of the subsequent regression estimations.

Table 2. Data Distribution by Industry

Industry Category	Number of Companies	Company Percentage (%)
Semiconductor	33	10.51%
Chemical Industry	32	10.19%
Electronic Components	27	8.60%
Computer & Peripheral Equipment	27	8.60%
Food Industry	19	6.05%
Telecommunications & Networking	18	5.73%
Others	17	5.41%
Optoelectronics	16	5.10%
Other Electronics	12	3.82%
Construction & Building Materials	12	3.82%
Biotechnology & Medical	11	3.50%
Textile & Fiber	11	3.50%
Plastics	10	3.18%
Steel Industry	9	2.87%
Automotive Industry	8	2.55%
Tourism & Hospitality	8	2.55%
Cement Industry	7	2.23%
Trading & Retail	7	2.23%
Electrical Machinery	7	2.23%
Shipping Industry	6	1.91%
Paper Manufacturing	4	1.27%
Electrical Appliances & Cables	3	0.96%
Rubber Industry	3	0.96%
Oil, Electricity & Gas	2	0.64%
Information Services	2	0.64%
Electronic Distribution	2	0.64%
Glass & Ceramics	1	0.32%
Total	314	100%

Table 3. Descriptive Statistics

Variables	Mean	Median	Max	Min	SD	N
Stage 1						
ER	2.27	-5.14	814.28	-91.92	45.10	2198
MKT	8.92	18.14	24.99	-28.23	19.32	2198
SMB	8.62	6.87	20.18	0.42	5.93	2198
HML	23.60	27.76	33.18	11.56	8.65	2198
PROF	0.10	0.09	0.88	-1.9	0.15	2198
INVEST	6.57	3.73	325.54	-51.05	17.36	2198
IDIRISK	-0.09	-0.23	15.30	-4.05	0.87	2198
Stage 2						
<i>Dependent</i>						
IDIRISK	-0.09	-0.23	15.30	-4.05	0.87	2198
<i>Independent</i>						
WI	0.33	0.03	153.50	0	4.97	2198
TC	4.16	3.25	43.06	0.52	3.36	2198
S1	4.01	3.08	41.23	0.42	3.26	2198
S2	4.20	3.25	42.58	0.37	3.46	2198
WIR	2.56	0	1910.83	0	55.42	2198
BS	9.37	9	21	5	2.30	2198
IDR	0.36	0.33	0.70	0.14	0.09	2198
FDR	0.17	0.15	0.58	0	0.11	2198
MSR	22.04	20.35	90.83	1.42	11.59	2198
MAR	0.81	0.14	14.62	0	1.75	2198
NEF	4.27	1	204	0	9.96	2198
NEI	0.02	0	5.41	0	0.15	2198
<i>Control</i>						
ESIZE	16.95	16.85	22.55	11.50	1.63	2198
ROA	5.59	5.02	61.07	-45.31	7.52	2198
FLEVE	2.10	1.80	19.97	1.02	1.19	2198
CASHR	0.16	0.14	0.72	0	0.10	2198
ASGR	-0.02	-0.02	1.79	-0.77	0.14	2198
REGR	0.05	0.03	3.03	-0.80	0.25	2198

Table 4. Pearson Correlation Matrix

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)
(a) IDIRISK	1																		
(b) WI	0.004	1																	
(c) TC	0.01	-0.03	1																
(d) S1	0.01	-0.03	0.99***	1															
(e) S2	-0.003	-0.04*	0.95***	0.97***	1														
(f) WIR	-0.02	-0.003	-0.02	-0.02	-0.01	1													
(g) BS	-0.08***	-0.01	0.03	0.02	0.01	-0.01	1												
(h) IDR	0.06***	-0.02	-0.02	-0.01	-0.002	0.04*	-0.70***	1											
(i) FDR	0.005	0.07***	-0.06***	-0.05**	-0.07***	0.01	-0.18***	0.17***	1										
(j) MSR	-0.01	0.07***	-0.67***	-0.67***	-0.66***	0.03	-0.07***	0.07***	0.17***	1									
(k) MAR	0.01	-0.02	0.03	0.05**	0.06***	0.08***	-0.05**	0.03	-0.01	-0.07***	1								
(l) NEF	-0.003	-0.01	-0.07***	-0.09***	-0.09***	-0.01	0.04**	-0.08***	-0.04*	0.09***	-0.07***	1							
(m) NEI	-0.00003	-0.002	-0.03	-0.03	-0.04*	-0.004	0.01	-0.06***	-0.01	0.05**	-0.01	0.12***	1						
(n) ESIZE	-0.01	-0.01	-0.06***	-0.10***	-0.09***	-0.01	0.21***	-0.02	-0.20***	0.04**	-0.19***	0.17***	-0.10***	1					
(o) ROA	0.11***	-0.01	-0.01	-0.01	-0.01	0.04*	0.03	0.04**	-0.05**	-0.04*	0.06***	-0.08***	-0.03	0.07***	1				
(p) FLEVE	-0.03	0.001	-0.05**	-0.05**	-0.05**	-0.01	0.04*	-0.04*	0.03	0.10***	-0.08***	0.11***	-0.03	0.31***	-0.21***	1			
(q) CASHR	0.10***	-0.06***	0.08***	0.10***	0.10***	0.02	-0.10***	0.16***	-0.0004	-0.15***	0.12***	-0.09***	-0.06***	-0.10***	0.22***	-0.19***	1		
(r) ASGR	0.06***	-0.001	-0.03	-0.03	-0.02	-0.05**	-0.05**	0.08***	0.08***	0.04*	-0.02	0.01	0.001	-0.04*	-0.25***	-0.04*	-0.03	1	
(s) REGR	0.18***	-0.02	0.01	0.01	0.01	0.01	0.002	-0.01	-0.01	-0.01	0.03	-0.02	-0.02	0.05**	0.32***	0.03	0.02	-0.13***	1

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 5. VIFs for the All Models

<i>Dependent</i>	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK
<i>Independent</i>												
WI	1.00											
TC		1.01										
S1			1.02									
S2				1.02								
WIR					1.00							
BS						1.06						
IDR							1.04					
FDR								1.06				
MSR									1.03			
MAR										1.05		
NEF											1.04	
NEI												1.01
<i>Control</i>												
ESIZE	1.14	1.14	1.14	1.14	1.14	1.18	1.14	1.19	1.14	1.17	1.16	1.15
ROA	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.32	1.33	1.32
FLEVE	1.21	1.21	1.21	1.21	1.21	1.21	1.21	1.22	1.21	1.21	1.21	1.21
CASHR	1.09	1.09	1.09	1.10	1.09	1.09	1.11	1.09	1.11	1.09	1.09	1.09
ASGR	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08	1.08
REGR	1.13	1.13	1.13	1.13	1.13	1.13	1.13	1.13	1.13	1.13	1.13	1.13

4.3 Model Selection

Tables 6–8 report 2SLS estimates, which are included to strengthen causal interpretation by addressing potential endogeneity. The validity of 2SLS depends on instruments that satisfy both relevance and exogeneity. In this dissertation, instrumental variables are specified for WI, TC, S1, S2, WIR, BS, IDR, FDR, MSR, MAR, NEF, and NEI, and instrument diagnostics are conducted. Across all models, the Cragg–Donald Wald F statistics exceed the Stock–Yogo 10% critical value of 16.38, indicating strong instruments and mitigating weak-instrument concerns. In addition, the Anderson LM test reports $p < 0.01$ throughout, indicating that under-identification is not a concern. Finally, a Sargan statistic of zero indicates exact identification, implying no over-identification issues.

4.4 Regression Analysis Results

Table 6 summarizes regression results for idiosyncratic risk under the environmental and social dimensions. For water intensity (WI), the WI coefficient turns positive and becomes statistically significant, which supports H1-A. This pattern indicates that heavier water use intensity may be linked to greater enterprise-specific risk exposure. Interpreted substantively, intensive water dependence may be perceived by the market as a recognizable risk-related signal.

However, total carbon emission intensity (TC) exhibits a significant relationship with idiosyncratic risk. Accordingly, H1-B is supported. In contrast, Scope 1 emission intensity (S1) shows no statistically significant results, providing no support for H1-C. The results for Scope 2 emission intensity (S2) are likewise insignificant, and thus H1-D is not supported.

Overall, among the environmental dimensions, WI exhibits a significantly positive effect, indicating that enterprises with lower water-use efficiency face higher idiosyncratic risk. This finding is consistent with the notion that high water intensity increases operational uncertainty

and regulatory pressure, which may be reflected in greater volatility in enterprise-specific returns. In addition, TC is positively and significantly associated with idiosyncratic risk, suggesting that firms with higher overall carbon emission burdens are exposed to greater firm-specific risk. In contrast, neither S1 nor S2 shows a statistically significant effect. These results indicate that, when assessing idiosyncratic risk, the market tends to focus on the long-term transition pressure and policy exposure reflected in enterprises' overall carbon emission levels rather than on the magnitude of emissions from individual sources.

Table 6. Empirical Results under the Environmental and Social Dimension Models

Methods	2SLS estimates				
	First Stage				
<i>Dependent</i>	WI	TC	S1	S2	WIR
<i>Constant</i>	1.49	-5.56	-5.78	-7.96	-163.82
<i>Industry_WI</i>	1.00***				
<i>Industry_TC</i>		0.85***			
<i>Industry_S1</i>			0.85***		
<i>Industry_S2</i>				0.88***	
<i>Lagged WIR</i>					-0.15***
<i>Control</i>	include	include	include	include	include
	Second Stage				
<i>Dependent</i>	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK
<i>Constant</i>	7.20***	8.18***	7.91***	7.43***	14.63***
<i>Independent</i>					
<i>WI (instrumented)</i>	0.04*				
<i>TC (instrumented)</i>		0.07*			
<i>S1 (instrumented)</i>			0.05		
<i>S2 (instrumented)</i>				0.01	
<i>WIR (instrumented)</i>					-0.003
<i>Control</i>					
<i>ESIZE</i>	-0.46***	-0.53***	-0.51***	-0.47***	-0.90***
<i>ROA</i>	0.02***	0.02***	0.02***	0.02***	0.02***
<i>FLEVE</i>	0.09**	0.10**	0.10**	0.10**	0.10*
<i>CASHR</i>	0.95***	0.91**	0.93**	0.94***	1.05**
<i>ASGR</i>	0.66***	0.70***	0.69***	0.68***	0.72***
<i>REGR</i>	0.56***	0.56***	0.56***	0.55***	0.74***
<i>CDWF</i>	188.31	122.68	121.06	138.30	48.02
<i>Stock-Yogo (10%)</i>	16.38	16.38	16.38	16.38	16.38
<i>UI</i>	171.78***	115.58***	114.06***	129.29***	46.80***
<i>Sargan statistic</i>	0	0	0	0	0
<i>N</i>	2198	2198	2198	2198	1884

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. CDWF stands for the Cragg–Donald Wald F statistic, while UI represents the under-identification test, reported using the Anderson canonical correlation LM statistic.

Table 6 also reports results for the social pillar using the workplace injury rate (WIR). The WIR coefficients is not statistically significant. Therefore, H2-A is not supported. This suggests that occupational safety information, at least as captured by WIR, does not generate a detectable effect on enterprises' idiosyncratic risk within the data period. One possible interpretation is that market reactions to employee safety performance are limited, or that improvements in occupational safety may require a longer horizon before being reflected in risk measures. Overall, the results indicate that the social indicator used here does not appear to be a central input in investors' idiosyncratic risk evaluation during the study period.

Table 7. Empirical Results under the Corporate Governance Dimension Models

Methods	2SLS estimates				
	First Stage				
<i>Dependent</i>	BS	IDR	FDR	MSR	MAR
<i>Constant</i>	-2.08	0.27**	-0.05	-2.85	1.71
Industry_BS	0.96***				
Industry_IDR		1.00***			
Industry_FDR			0.85***		
Lagged_MSR				0.43***	
Lagged_MAR					0.47***
<i>Control</i>	include	include	include	include	include
	Second Stage				
<i>Dependent</i>	IDIRISK	IDIRISK	IDIRISK	IDIRISK	IDIRISK
<i>Constant</i>	8.41***	7.23***	6.76***	14.09***	14.87***
<i>Independent</i>					
BS (instrumented)	-0.12**				
IDR (instrumented)		-0.02			
FDR (instrumented)			-2.28*		
MSR (instrumented)				0.03**	
MAR (instrumented)					0.05
<i>Control</i>					
ESIZE	-0.46***	-0.46***	-0.41***	-0.03***	-0.91***
ROA	0.02***	0.02***	0.02***	0.02***	0.02***
FLEVE	0.08*	0.09**	0.08*	0.10	0.10*
CASHR	1.03***	0.94***	0.93**	1.06**	1.07**
ASGR	0.68***	0.68***	0.77***	0.74***	0.77***
REGR	0.56***	0.55***	0.54***	0.72***	0.73***
CDWF	300.46	621.86	277.29	382.67	440.16
Stock-Yogo (10%)	16.38	16.38	16.38	16.38	16.38
UI	259.96***	468.85***	242.50***	308.79***	344.98***
Sargan statistic	0	0	0	0	0
N	2198	2198	2198	1884	1884

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. CDWF stands for the Cragg–Donald Wald F statistic, while UI represents the under-identification test, reported using the Anderson canonical correlation LM statistic.

Table 7 summarizes how governance variables relate to IDIRISK, distinguishing between board characteristics and ownership structure. For board size (BS), the estimate shows a significantly negative coefficient, indicating that enterprises with larger boards experience lower enterprise-specific risk; this finding supports H3-A. A plausible interpretation is that larger boards bring broader expertise and more extensive oversight capacity, which can strengthen monitoring, reduce informational frictions, and dampen operational uncertainty.

In contrast, IDR is not statistically significant; thus, H3-B is not supported. This outcome implies that the presence of independent directors, while intended to enhance monitoring, does not translate into measurable changes in enterprise-specific risk in this dataset. One possible explanation is that independent directors may face practical constraints that limit their ability to materially influence strategic decisions or risk governance, weakening any risk-mitigating effect.

However, the estimated coefficient on FDR is significantly negative, indicating that enterprises with a higher proportion of female directors exhibit significantly lower idiosyncratic risk. Accordingly, Hypothesis H3-C is supported. This finding suggests that board gender diversity may help reduce enterprise-level uncertainty. By incorporating diverse perspectives and decision-making approaches, gender-diverse boards are likely to enhance overall board effectiveness and risk oversight, which is reflected in the reduction of idiosyncratic risk.

Turning to ownership structure, MSR becomes positive and statistically significant, thereby supporting H3-D. This result indicates that greater ownership concentration is associated with higher enterprise-specific risk. A possible interpretation is that concentrated control may encourage more aggressive strategies or stronger risk-taking preferences among controlling owners, raising operational uncertainty. The finding also implies that concentrated ownership does not automatically function as a risk-reducing monitoring device; without effective countervailing oversight, it may instead intensify moral hazard and facilitate outcomes consistent with a “tunneling” logic.

Finally, MAR is not statistically significant, meaning H3-E is not supported. This suggests that aligning ownership and control through managerial shareholding does not necessarily produce observable risk-governance improvements. The effect of managerial ownership may be contingent on enterprise-level institutional settings and incentive design, which may not be captured in idiosyncratic risk over the data horizon.

Table 8. Empirical Results under the ESG-Related Negative Event Models

Methods	2SLS estimates	
	First Stage	
<i>Dependent</i>	NEF	NEI
<i>Constant</i>	-6.03	-0.32
Industry_NEF	1.00***	
Industry_NEI		0.99***
<i>Control</i>	include	include
	Second Stage	
<i>Dependent</i>	IDIRISK	IDIRISK
<i>Constant</i>	7.22***	7.17***
<i>Independent</i>		
NEF (instrumented)	0.0005	
NEI (instrumented)		-0.27
<i>Control</i>		
ESIZE	-0.46***	-0.46***
ROA	0.02***	0.02***
FLEVE	0.09**	0.09**
CASHR	0.94***	0.93***
ASGR	0.68***	0.68***
REGR	0.55***	0.55***
CDWF	352.23	193.00
Stock-Yogo (10%)	16.38	16.38
UI	297.68***	175.66***
Sargan statistic	0	0
N	2198	2198

Notes: *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. CDWF stands for the Cragg–Donald Wald F statistic, while UI represents the under-identification test, reported using the Anderson canonical correlation LM statistic.

Overall, Table 7 shows that BS and FDR are the governance indicators with a consistent, robust association with risk reduction. By comparison, only IDR shows no statistically detectable effects. For ownership variables, the evidence is mixed: MSR becomes idiosyncratic

risk increasing, whereas MAR remains insignificant. Taken together, these results suggest that board monitoring capacity and governance effectiveness appear more influential for idiosyncratic risk than ownership concentration alone. In this setting, strengthening board oversight is more directly linked to idiosyncratic risk mitigation, while ownership concentration may require strong external checks to prevent excessive risk-taking by large shareholders.

Table 8 examines whether disclosures of ESG-related adverse events, captured by event frequency (NEF) and event severity (NEI), are associated with idiosyncratic risk. Neither NEF nor NEI is statistically significant, indicating no measurable relationship between these negative-event indicators and enterprise-specific risk during the data period. Accordingly, H4-A and H4-B are not supported.

Substantively, while ESG controversies may harm reputation and influence short-term sentiment, the results imply that their frequency and measured severity do not translate into persistent enterprise-specific risk as captured by IDIRISK. One interpretation is that negative ESG news is more likely to be reflected in transitory price reactions rather than in a sustained increase in fundamental operational uncertainty. The risk relevance of ESG negative events may depend on contextual conditions, such as the nature of the incident, the degree of media amplification, and the enterprise's response, rather than being captured solely by annual counts or fine-based intensity measures.

4.5 Discussion

This dissertation evaluates how sustainability performance across the environmental (E), social (S), and governance (G) pillars, together with ESG-related negative event disclosures, relates to enterprise risk. Emphasis is placed on how external sustainability information interacts with internal governance mechanisms in shaping investors' risk perceptions. The hypothesis testing summary (Table 9) points to a clear theme: ESG effects are heterogeneous,

and enterprise risk does not appear to be driven by a single, uniform ESG factor. Instead, the market seems to respond selectively to particular ESG signals.

From the environmental perspective, investors do not react evenly across indicators. Once endogeneity is addressed, water intensity (WI) shows a significantly positive relationship with idiosyncratic risk, implying that water dependence is perceived as an enterprise-specific vulnerability. This is consistent with the idea that water-related exposure, such as supply instability, regulatory constraints, or operational disruptions, can be highly idiosyncratic and thus priced as a meaningful risk signal.

Table 9. Research Hypothesis Testing Results

Dimension	Statement	Testing Results
Environmental	<i>H1-A: WI is positively related to IDIRISK.</i>	Y
	<i>H1-B: TC is positively related to IDIRISK.</i>	Y
	<i>H1-C: S1 is positively related to IDIRISK.</i>	N
	<i>H1-D: S2 is positively related to IDIRISK.</i>	N
Social	<i>H2-A: WIR is positively related to IDIRISK.</i>	N
Corporate Governance	<i>H3-A: BS is negatively related to IDIRISK.</i>	Y
	<i>H3-B: IDR is negatively related to IDIRISK.</i>	N
	<i>H3-C: FDR is negatively related to IDIRISK.</i>	Y
	<i>H3-D: MSR is positively related to IDIRISK.</i>	Y
	<i>H3-E: MAR is negatively related to IDIRISK.</i>	N
ESG-Related Negative	<i>H4-A: NEF is positively related to IDIRISK.</i>	N
Events	<i>H4-B: NEI is positively related to IDIRISK.</i>	N

Note: Y denotes hypothesis supported; N denotes hypothesis not supported.

Similarly, TC is positively and significantly associated with idiosyncratic risk, suggesting that enterprises with higher overall carbon burdens are perceived as facing elevated transition-

and policy-related risks. In contrast, neither S1 nor S2 displays a statistically significant effect. This divergence implies that investors do not appear to assess idiosyncratic risk based on emissions from individual sources in isolation. Instead, market participants focus on enterprises' aggregate carbon exposure, which better captures long-term transition pressures, regulatory risks, and strategic adaptation challenges. Taken together, these results suggest that holistic environmental exposure, rather than granular emission components, is more relevant for explaining variations in idiosyncratic risk.

For the social pillar, WIR does not show a statistically significant association with enterprise risk in either estimation approach. This suggests that occupational safety information has not yet become a primary input in investors' risk evaluation during the data period. Social improvements may accumulate slowly and be less immediately visible in risk measures, and investors may still treat social responsibility largely as a compliance domain rather than a direct driver of operational volatility. This interpretation aligns with strands of the ESG literature that observe weaker pricing effects for social indicators relative to environmental and governance measures, implying that the social dimension may still be at an earlier stage of market materialization.

The governance results are more consistent and pronounced. BS is robustly and negatively related to idiosyncratic risk, indicating that larger boards are associated with lower enterprise-specific risk. This pattern accords with agency-based arguments and monitoring views: broader expertise and collective oversight can strengthen monitoring quality, reduce information asymmetry, and constrain managerial opportunism, thereby lowering risk.

Board gender diversity also emerges as a key mechanism for reducing idiosyncratic risk. The significantly negative coefficient on FDR suggests that gender-diverse boards are associated with lower idiosyncratic risk. This result aligns with arguments that gender diversity improves deliberation quality, risk awareness, and governance effectiveness, thereby contributing to more prudent decision-making and enhanced idiosyncratic risk control. By

contrast, IDR does not exhibit a statistically significant effect. This outcome implies that formal independence alone may be insufficient to materially influence idiosyncratic risk outcomes, potentially due to institutional constraints or limited involvement in strategic risk decisions.

Finally, ownership variables show divergent patterns. MAR is not significant, indicating that equity-based alignment does not automatically produce measurable risk reductions. However, MSR becomes significantly positive after endogeneity is controlled, implying that large shareholders' ownership concentration can increase idiosyncratic risk. This result is consistent with wealth-transfer arguments: when dominant shareholders are insufficiently constrained, they may pursue strategies that elevate enterprise-specific uncertainty and risk.

With respect to ESG-related negative event disclosures, the empirical results indicate that neither NEF nor NEI has a significant impact on enterprises' idiosyncratic risk. This finding suggests that the market may perceive ESG controversies primarily as short-term or reputational shocks rather than as persistent sources of operational risk. Investors may place greater emphasis on enterprises' response capabilities, disclosure practices, and crisis management strategies than on the occurrence or severity of negative events.

Overall, the findings indicate that investors' evaluations of sustainability information and enterprise risk are highly selective and context-dependent. Environmental risks that are closely tied to enterprises' operations (such as water intensity) and governance mechanisms that reflect substantive monitoring capacity (such as board size and board gender diversity) are more likely to be incorporated into risk pricing. When ownership concentration among large shareholders becomes excessive, it may enable them to exert greater influence over corporate decision-making, leading enterprises to adopt strategies that favor the private benefits of large shareholders and increase idiosyncratic risk. Taken together, enterprise risk formation appears to rely more on governance quality and specific operational constraints than on the sheer

intensity of ESG disclosures, underscoring the importance of substance over form in sustainability-related risk assessment.

Further analysis indicates that the key determinants of idiosyncratic risk are primarily derived from enterprise-disclosed information, suggesting that the information generated by different ESG dimensions is heterogeneous and should therefore be interpreted separately. Aggregating all ESG dimensions into a single composite measure may obscure important risk-relevant information and may also incorporate variables that do not materially affect idiosyncratic risk, thereby leading to inefficient resource allocation. According to the empirical results, the ESG dimensions that exert significant effects on idiosyncratic risk mainly include water intensity and total carbon emission intensity within the environmental dimension, as well as board size, board gender diversity, and large shareholder ownership within the corporate governance dimension. In contrast, the disclosure of negative ESG-related information exhibits a relatively limited impact on idiosyncratic risk, indicating that idiosyncratic risk is influenced primarily by voluntarily disclosed enterprise-level information.

V. Conclusion, Research Limitations, and Future Research Directions

5.1 Conclusion

This dissertation systematically examines how enterprises' sustainability performance across the environmental (E), social (S), and corporate governance (G) dimensions, as well as ESG-related negative event disclosures, affect enterprise risk. By employing the 2SLS approach, this dissertation enhances the rigor of causal inference and effectively addresses potential endogeneity concerns.

Overall, within the environmental dimension, WI and TC emerge as significant drivers of idiosyncratic risk. Enterprises with lower water-use efficiency and higher overall carbon emission burdens exhibit greater idiosyncratic risk, suggesting that markets perceive intensive resource dependence and aggregate carbon exposure as salient signals of operational uncertainty, regulatory pressure, and long-term transition risk. In contrast, disaggregated carbon emission measures, S1 and S1, do not display significant effects, indicating that investors focus more on enterprises' overall carbon profiles than on individual emission sources when evaluating idiosyncratic risk.

Regarding the social dimension, occupational safety performance, as measured by WIR, does not exert a significant influence on enterprise risk. This result indicates that social dimension information has not yet become a core basis for investors' risk evaluation. It suggests that investors may continue to regard social responsibility primarily as a compliance requirement rather than a key determinant of enterprises' operational uncertainty, or that the risk effects of social performance require long-term accumulation to be gradually reflected in investment valuations.

By contrast, the corporate governance dimension exhibits the strongest explanatory power for enterprise risk. BS and IDR are both associated with lower idiosyncratic risk, highlighting the importance of board monitoring capacity and diversity in mitigating firm-specific

uncertainty. In contrast, the proportion of independent directors does not show a significant effect, suggesting that formal independence alone may be insufficient to influence risk outcomes.

With respect to ownership structure, MAR does not have a significant effect on enterprise risk, indicating that equity-based incentives do not necessarily function as effective risk-governance mechanisms. In contrast, MSR exhibits a significantly positive relationship with idiosyncratic risk, suggesting that ownership concentration may increase enterprises' operational uncertainty by amplifying large shareholders' risk-taking preferences or tunneling incentives.

Regarding ESG-related negative event disclosures, the empirical results show that neither NEF nor NEI has a statistically significant effect on enterprises' idiosyncratic risk. This finding suggests that while ESG-related negative events may attract investor attention or induce short-term valuation fluctuations, their effects have not been translated into significant changes in enterprises' internal operational uncertainty. In other words, investors may perceive ESG-related negative events as temporary or exogenous information shocks rather than as critical factors capable of altering enterprises' fundamental risk structures.

This result further implies that the impact of ESG-related negative events on enterprise risk may depend heavily on contextual factors such as event characteristics, media exposure, and enterprises' response strategies, rather than being determined solely by event frequency or penalty severity. Moreover, delays in the disclosure of negative event information may lead investors' risk assessments to form earlier, making long-term risk effects difficult to capture empirically. Overall, ESG-related negative event disclosures have not yet formed a stable or identifiable transmission mechanism affecting enterprises' idiosyncratic risk.

Taken together, the findings indicate that idiosyncratic risk is reduced primarily by environmental and corporate governance dimension performance. Board size and large shareholder ownership structure serve as key corporate governance mechanisms affect

idiosyncratic risk. Accordingly, enterprises should not only enhance the quality of sustainability disclosures, but also strengthen board governance structures, carefully manage large shareholder ownership concentration risks, and establish robust reputation and crisis management mechanisms to mitigate investors' negative risk perceptions.

Regarding carbon emission intensity, the empirical results indicate a significantly positive relationship between enterprises' overall carbon emission intensity and idiosyncratic risk, suggesting that aggregate carbon burdens have become an important consideration for investors when assessing idiosyncratic risk. In contrast, neither S1 nor S2 shows a statistically significant effect. This finding implies that market assessments of carbon-related risk tend to emphasize the long-term transition pressure and policy exposure reflected in enterprises' overall carbon profiles rather than the magnitude of emissions from individual sources. Accordingly, enterprises' carbon management strategies should avoid focusing solely on improvements in specific emission categories and instead adopt a more comprehensive approach that addresses overall operational structures and energy-use configurations. Systematically reducing total carbon emission intensity may help alleviate potential transition risks and regulatory uncertainty.

More specifically, enterprises may reduce overall carbon intensity by promoting low-carbon process transformation, improving energy-use efficiency, adjusting product portfolios, or gradually increasing the share of renewable energy consumption. Compared with symbolic carbon disclosures or isolated emission-reduction initiatives, such structural decarbonization measures are more likely to reduce enterprises' long-term carbon risk exposure and may ultimately be reflected in lower levels of idiosyncratic risk in capital markets.

From a corporate governance perspective, enterprises can enhance risk management by establishing an appropriate board size and promoting board gender diversity, thereby strengthening monitoring capacity and decision-making diversity. By maintaining a suitably sized board and appointing directors with diverse professional backgrounds and experience,

enterprises can improve the board's overall effectiveness in risk assessment, strategic deliberation, and oversight of managerial decision-making. In particular, greater diversity in directors' backgrounds and perspectives may help prevent dominance by a single line of thinking in the decision-making process, thereby reducing enterprise-level uncertainty.

Furthermore, in response to excessive ownership concentration among large shareholders, enterprises should adopt complementary mechanisms to mitigate their influence on managerial decision-making. Specifically, enterprises may achieve a more dispersed ownership structure through mechanisms such as issuing new equity, attracting institutional investors, encouraging large shareholders to divest part of their holdings, and implementing employee stock ownership plans. These measures can effectively constrain the influence of large shareholders and reduce idiosyncratic risk arising from the pursuit of private benefits.

5.2 Research Limitations and Future Research Directions

Although this dissertation provides empirical evidence on the relationship between ESG performance and enterprise risk, several limitations remain, which also offer directions for future research. First, the ESG indicators employed in this study are primarily derived from enterprises' publicly disclosed information. However, substantial variation exists across enterprises in terms of disclosure depth, reporting formats, and data reliability. Moreover, Taiwan's sustainability disclosure framework is still in a developmental stage, which may influence the empirical results for the environmental and social dimensions. As disclosure regulations become more comprehensive, future research may observe different relationships between ESG performance and enterprise risk. Second, disclosures of ESG-related negative events and penalties often involve reporting delays, while investors' risk assessments may form at an early stage of the event. This temporal mismatch may lead empirical models to underestimate the impact of negative events on enterprise risk, particularly with respect to idiosyncratic risk. Third, the data in this dissertation are limited to Taiwan's listed enterprises,

and the specific institutional environment, ownership structure, and corporate governance characteristics may constrain the external generalizability of the findings. Future research could adopt cross-country or cross-institutional comparative analyses to further examine whether the effects of ESG performance on enterprise risk differ across regulatory regimes and cultural contexts. Finally, this study primarily employs quantitative regression methods, which are effective in capturing systematic relationships among variables but may not fully reflect qualitative factors such as boardroom dynamics, organizational culture, or internal decision-making processes. As a result, certain risk formation mechanisms may not be fully captured in the empirical models.

Building on these limitations, future research may proceed along several promising avenues. As international sustainability disclosure frameworks, such as SASB, TCFD, and the EU's CSRD, continue to converge with Taiwan's regulatory environment, subsequent studies may adopt more standardized measurement indicators and integrate textual analysis or machine learning techniques to enhance the depth and scalability of ESG information analysis. With respect to ESG-related negative events, future research may further differentiate event types and incorporate factors such as media exposure intensity, information diffusion speed, and enterprises' response strategies to more precisely capture the risk transmission mechanisms associated with sustainability controversies. In addition, cross-country comparative studies and industry-level analyses may help deepen understanding of how institutional differences shape ESG risk pricing behavior. As digital technologies and AI-based regulatory tools continue to develop rapidly, future research may also explore how digitalization enhances ESG information transparency, improves risk identification efficiency, and strengthens corporate sustainability governance, thereby more comprehensively reflecting global trends in sustainability governance.

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